알레르기 혈액검사, 해석과 설명

2026.05.30 면역전문임상병리사 Workshop

세브란스병원 알레르기내과

오현경

Contents

Concept of Allergy

Clemens von Pirquet (1906)

Allergy = Change in reactivity

Discovery of Reagin

Carl Prausnitz & Heinz Küstner (1921)

Reagin = Ab-like substance responsible for induction of the erythema-wheal reaction

Identification of IgE

Kimishige Ishizaka & Teruko Ischizaka (1966)

Reagin = IgE

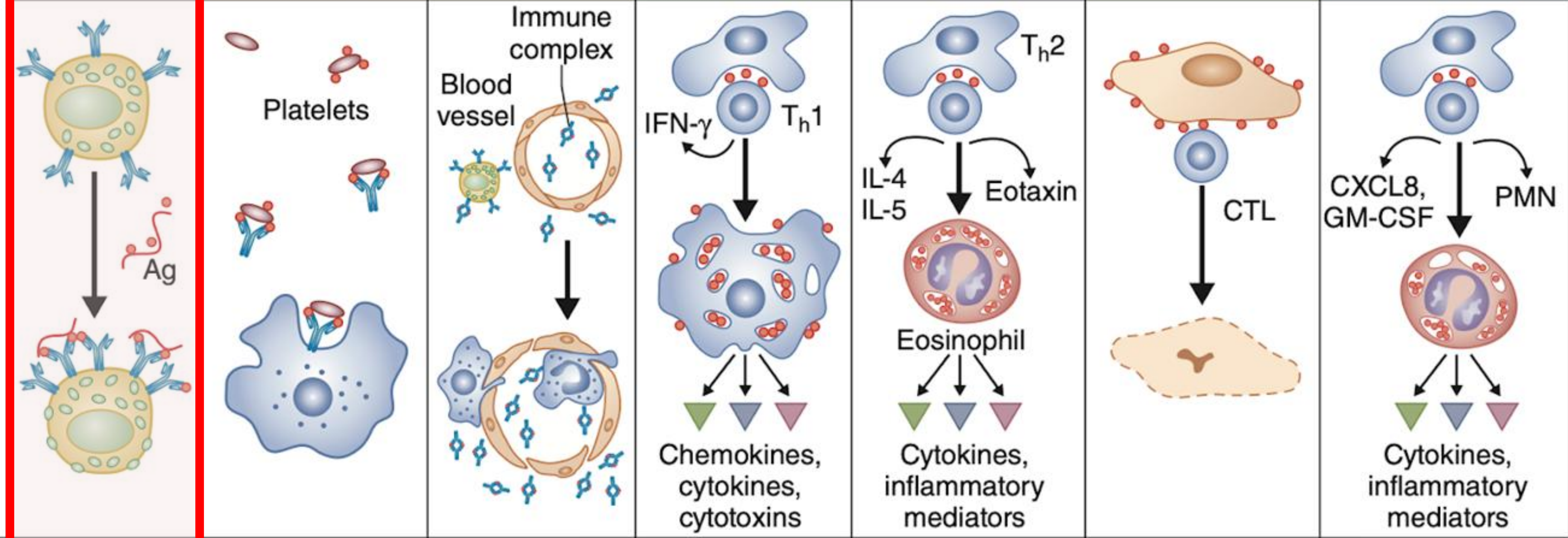
→ *in vitro* 검사 가능해짐

Igea JM. Allergy. 2013 Aug;68(8):966-73.

Ishizaka K, Ishizaka T. J Allergy Clin Immunol. 2016 Jun;137(6):1646-1650

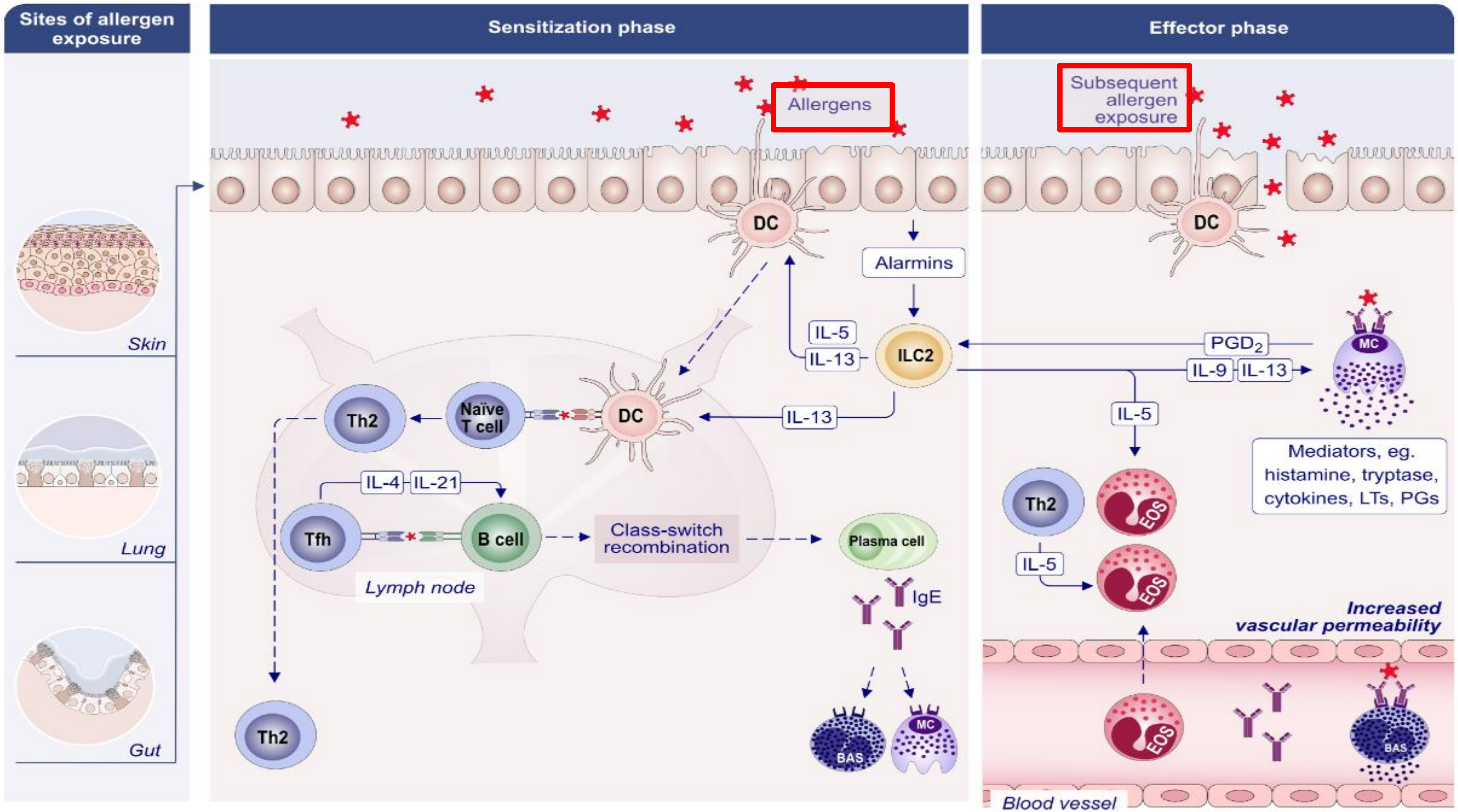
≫ 알레르기 병인기전

	Type I	Type II	Type III	Type IVa	Type IVb	Type IVc	Type IVd
Immune reactant	IgE	IgG	IgG	IFN-γ, TNF-α (T_h1 cells)	IL-5, IL-4/IL-13 (T_h2 cells)	Perforin/ granzyme B (CTL)	CXCL8, GM-CSF (T cells)
Antigen	Soluble antigen	Cell- or matrix-associated antigen	Soluble antigen	Antigen presented by cells or direct T-cell stimulation	Antigen presented by cells or direct T-cell stimulation	Cell-associated antigen or direct T-cell stimulations or direct T-cell stimulation	Soluble antigen presented by cells or direct T-cell stimulation
Effector	Mast cell activation	FcR+ cells (phagocytes, NK cells)	FcR+ cells Complement	Macrophage activation	Eosinophils	T cells	Neutrophils
Example of hypersensitivity reaction	Allergic rhinitis, sytemic anaphylaxis	Hemolytic anemia, thrombocyto-penia (e.g., penicillin)	Serum sickness, Arthus reaction	Tuberculin reaction, contact dermatitis (with IVc)	DRESS Maculopapular exanthema with eosinophilia	SJS/TEN Bullous exanthema and fixed drug eruption Hepatitis	AGEP Behçet disease



Burks AW. Et al. Middleton's Allergy: Principles and Practice (9th ed.), Amsterdam, Elsevier; 2020.

» 알레르기 병인기전



Jutel M Agache I et al. Allergy. 2023;78(11):2851-2874.

» 국내 주요 알레르겐

흡입 알레르겐

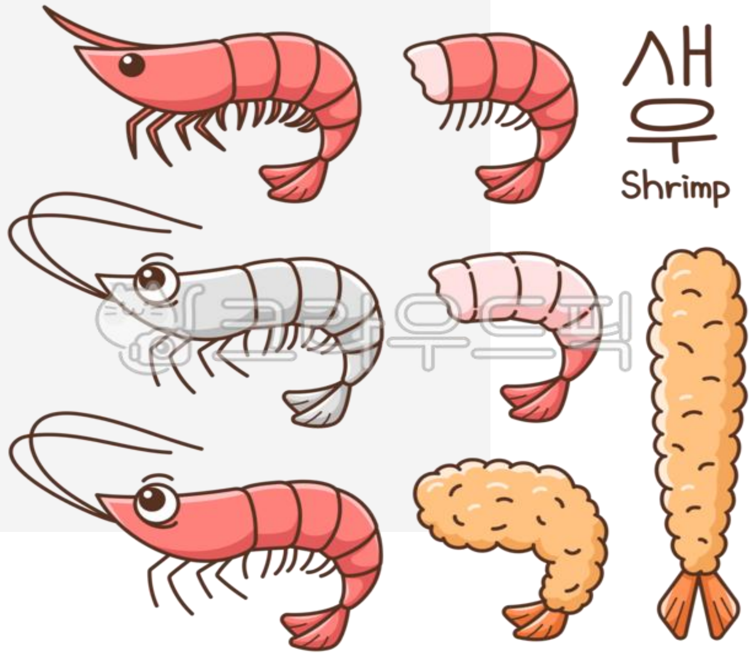
Outdoor allergen

Indoor allergen

식품 알레르겐

소아

성인



Indoor allergen

- 집먼지 진드기 (*Dermatophagoides farina, D. pteronyssinus*)
- 반려동물 (고양이, 강아지)
- 바퀴벌레 (*Blatella Germanica, Periplanta americana*)
- 실내 곰팡이 (*Aspergillus, Penicillium*)

Outdoor allergen

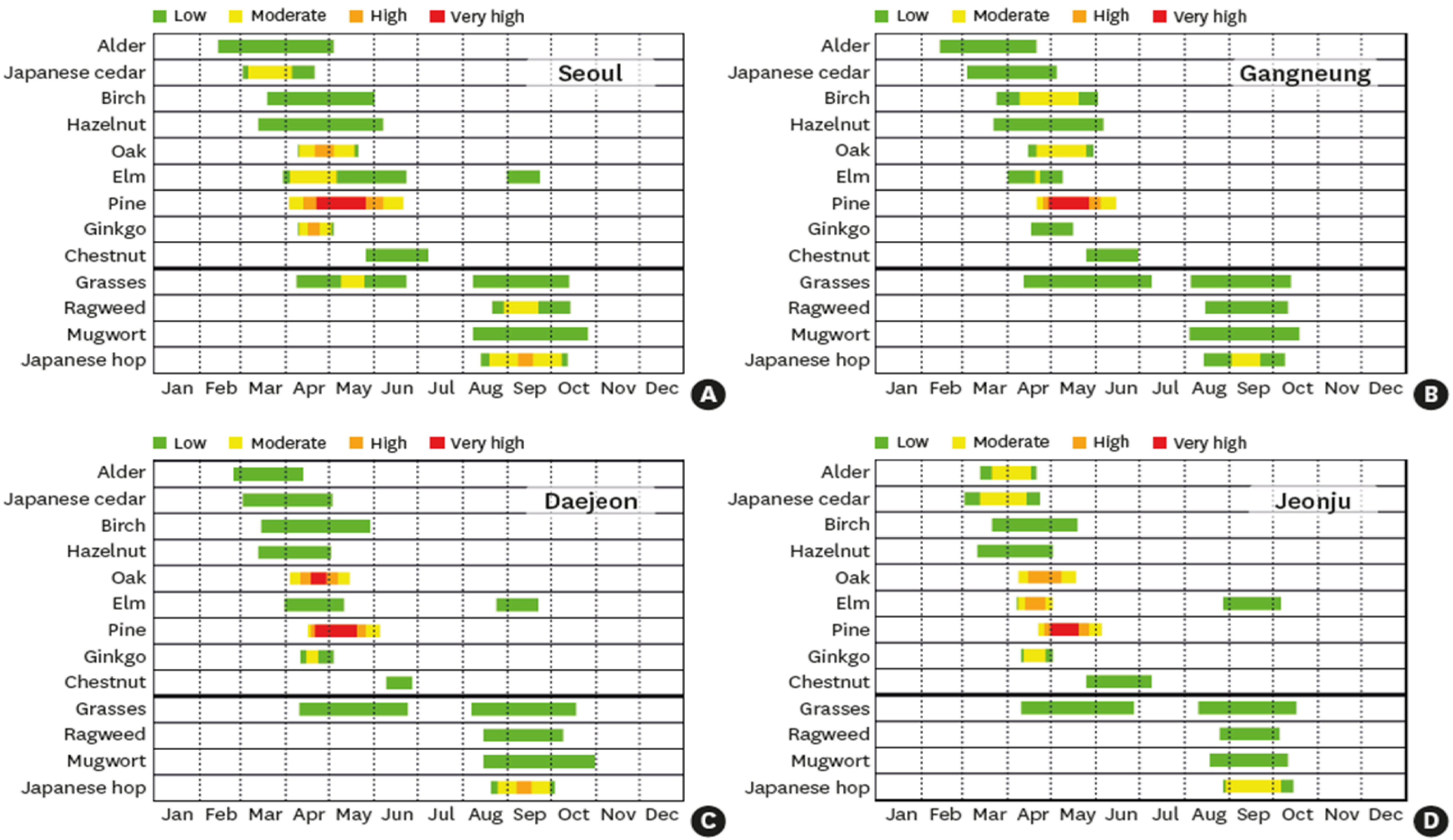
- 봄철 Tree pollen - 자작나무 (Birch), 참나무 (Oak), 오리나무 (Alder)
- 초여름 Grass pollen –큰조아재비 (Timothy), 호밀풀 (Rye grass), 우산잔디 (Bermuda)
- 가을철 Weed pollen – 쑥 (Mugwort), 돼지풀 (Ragweed), 환삼덩굴 (Hop Japanese)
- 실외 곰팡이 (Alternaria, Cladosporium)

국내 주요 알레르겐 _흡입 알레르겐

봄 : Tree pollens

여름 : Grass pollens

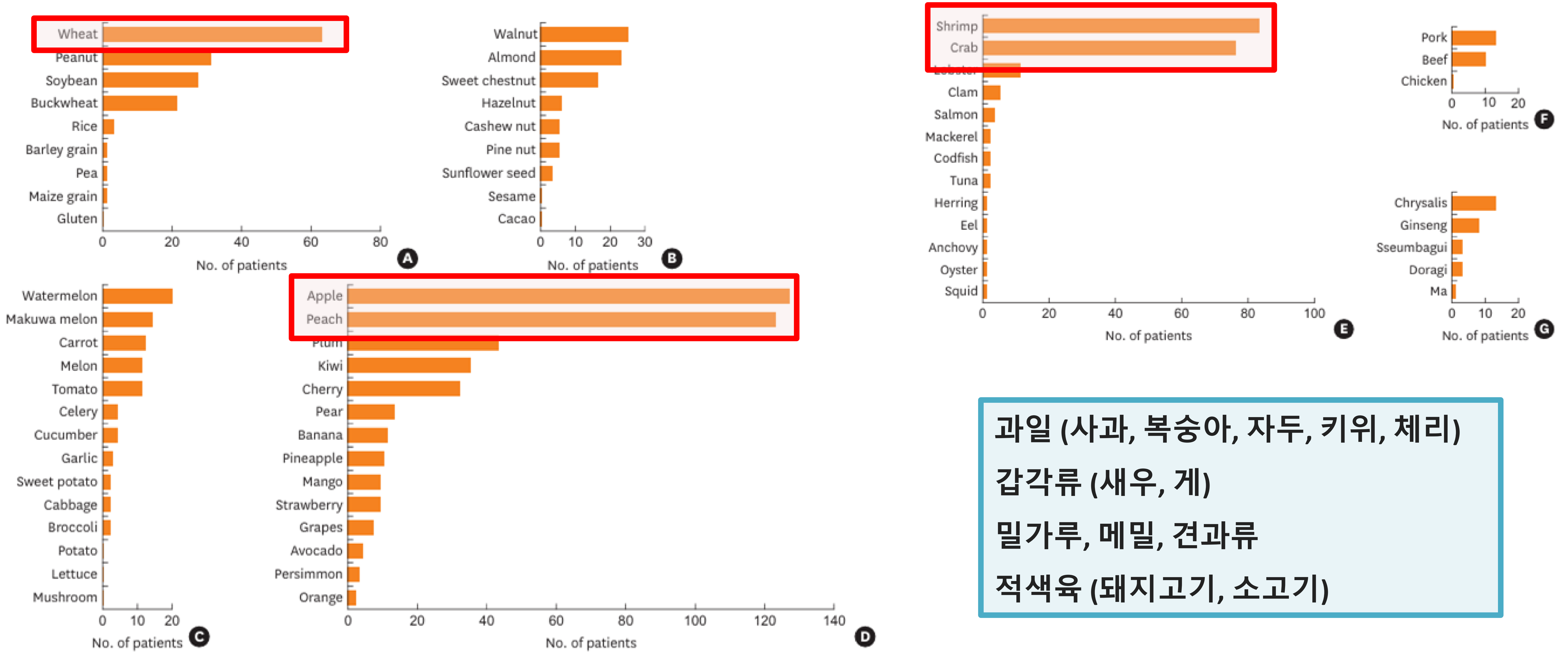
가을 : Weed pollens



Shin JY et al. Allergy Asthma Immunol Res. 2020;12(2):259-273.

국내 주요 알레르겐 _식품 알레르겐

국내 주요 알레르겐 _식품 알레르겐



과일 (사과, 복숭아, 자두, 키위, 체리)

갑각류 (새우, 게)

밀가루, 메밀, 견과류

적색육 (돼지고기, 소고기)

Lee SC et al. Allergy Asthma Immunol Res. 2019;11(5):723-735.

진단

History taking, P/E

In vivo test

In vitro test

치료

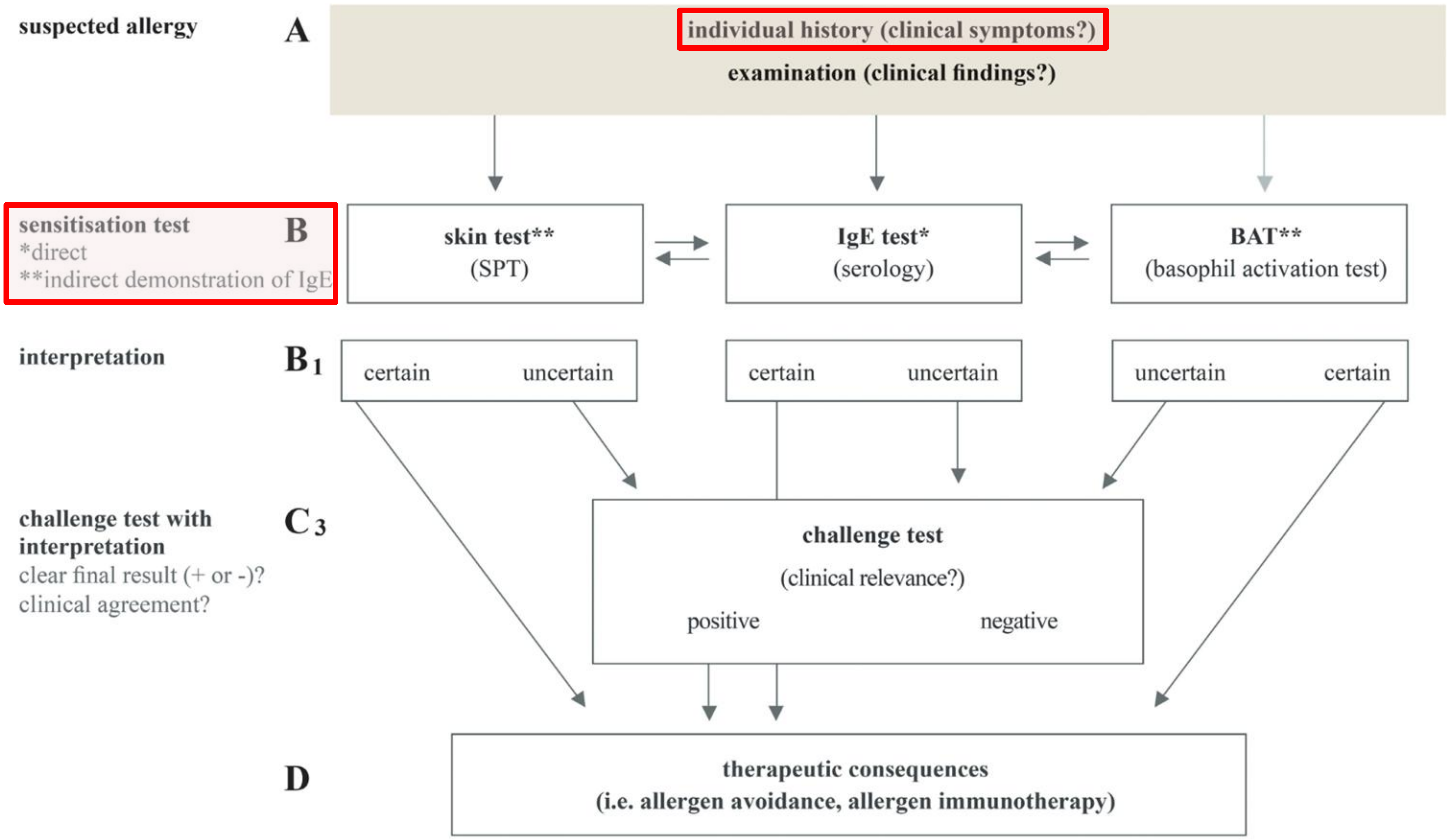
회피 요법

약물 요법

알레르겐 면역치료

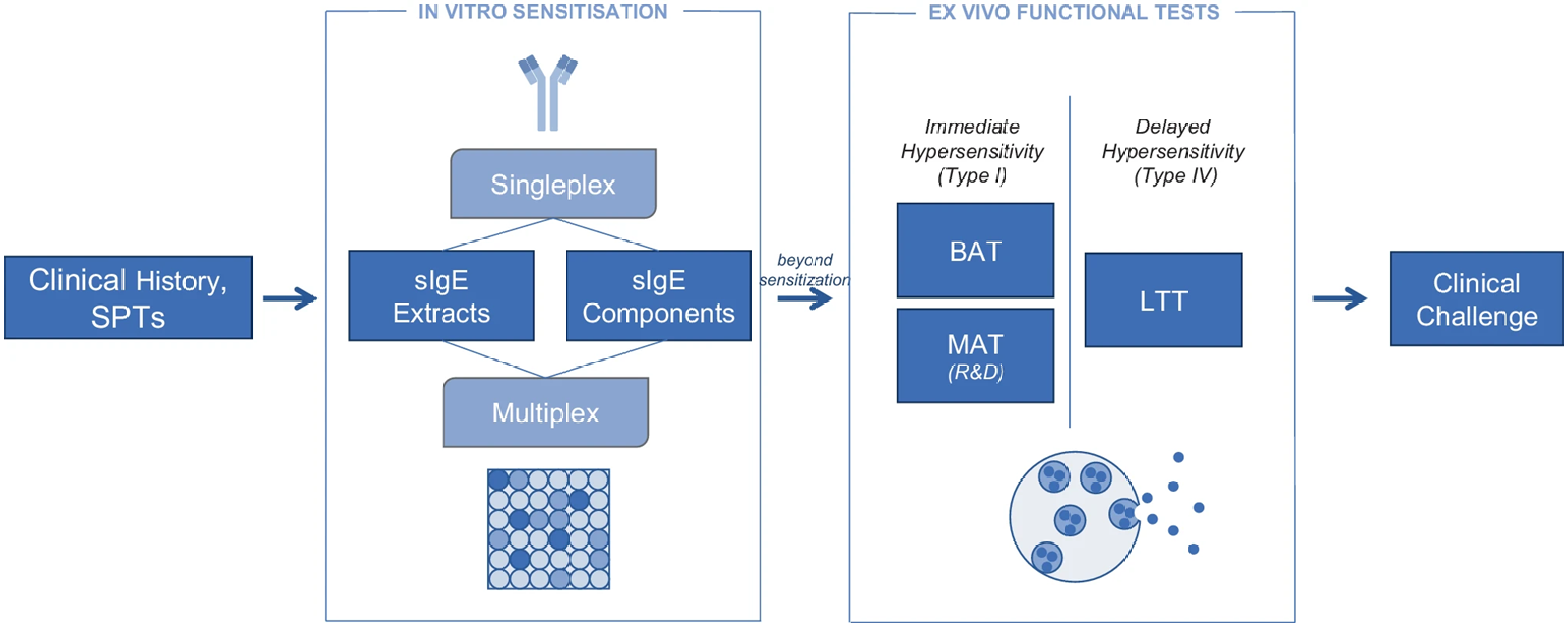
예방

증상 예측 및 대처



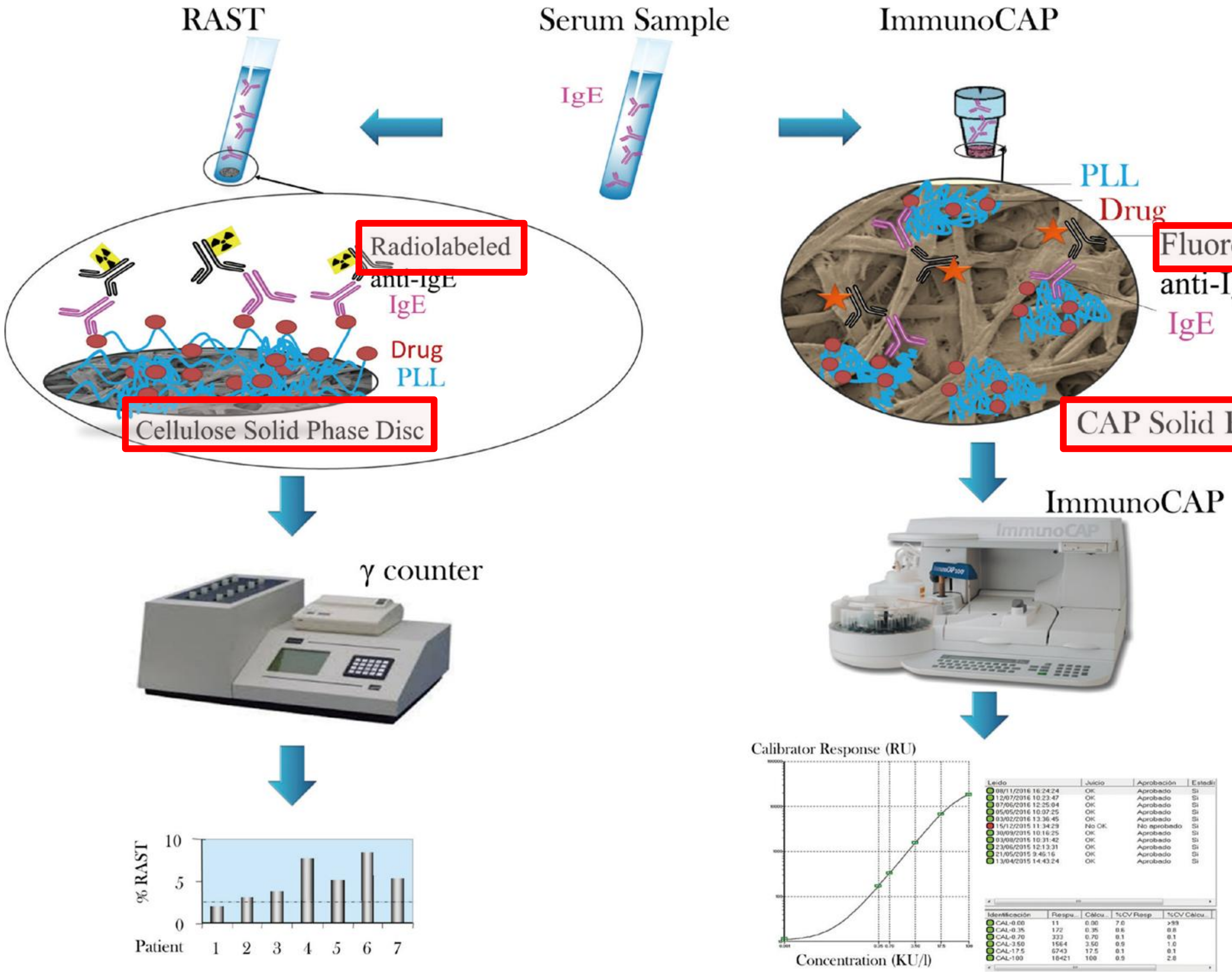
Dramburg S et al. Pediatr Allergy Immunol. 2023 Mar;34 Suppl 28:e13854.

» 알레르기 혈액 검사의 종류



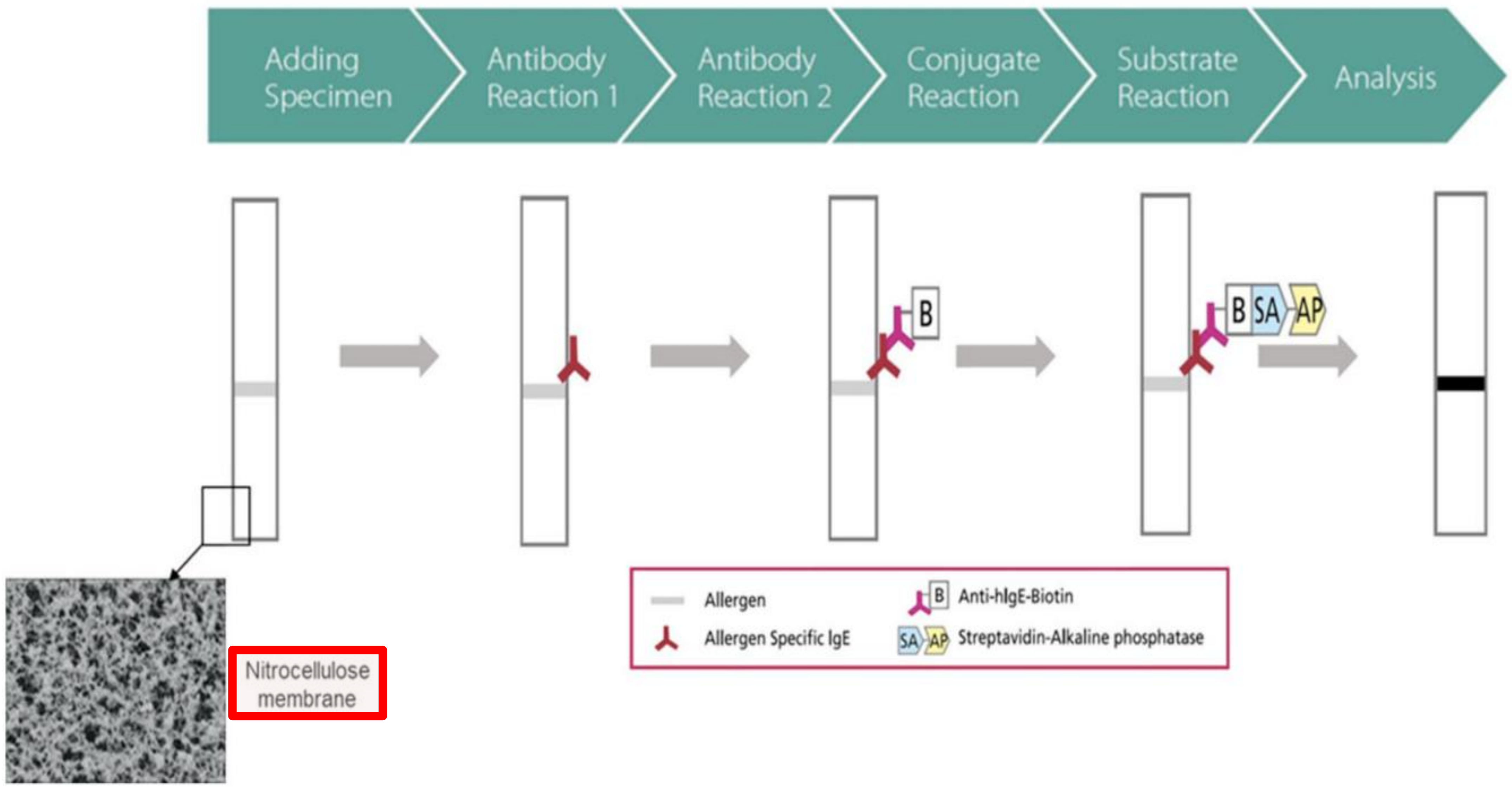
Arsenis C et al. Allergo J Int 34, 21–30 (2025).

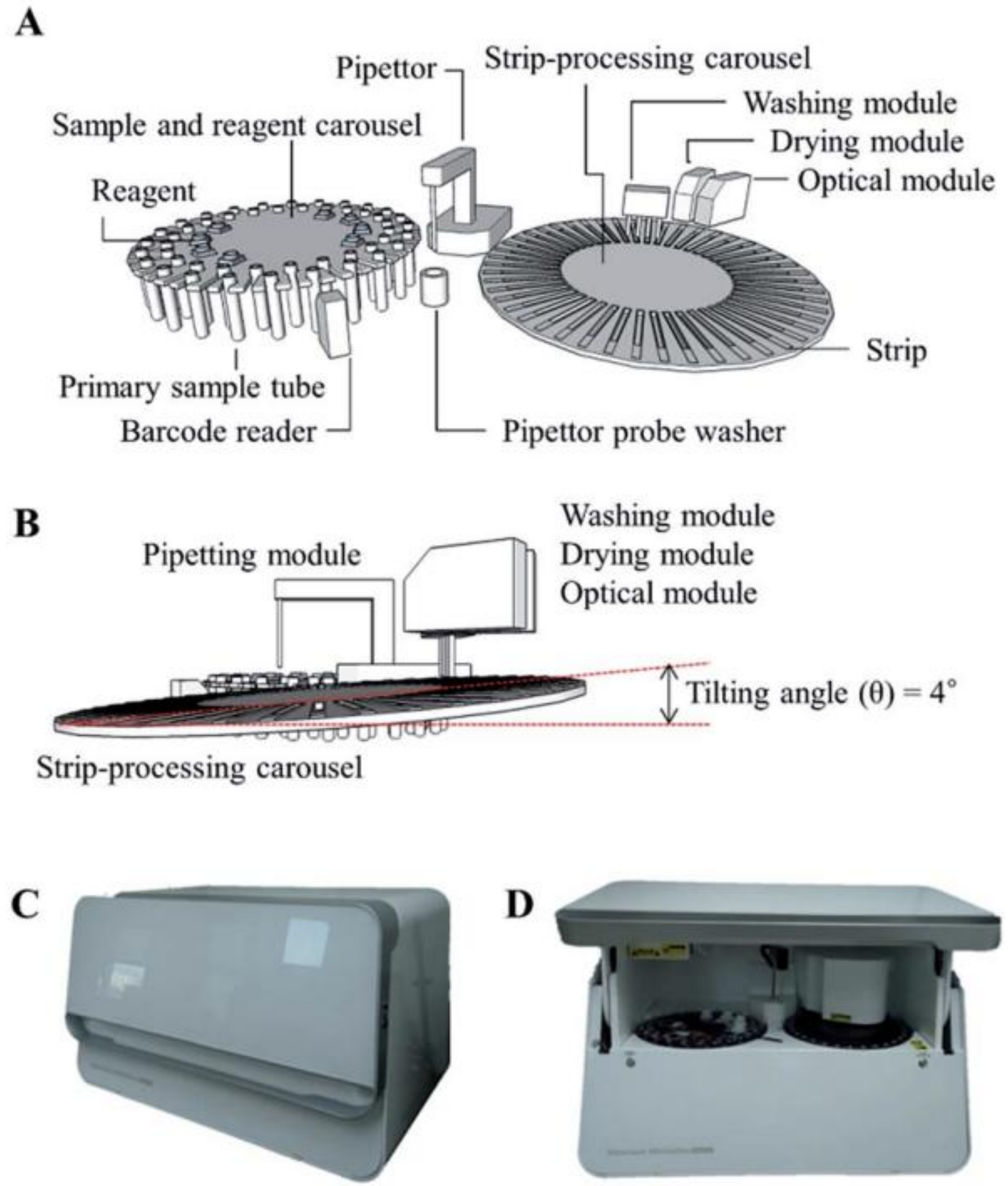
알레르기 혈액 검사의 종류_ImmunoCAP



Doña I et al. Allergy Asthma Immunol Res. 2017;9(4):288-298.

MAST
(Multiple Allergen Simultaneous Test)





Han JY et al. J. Clin. Med. 2021, 10, 960.
Oh JH et al. Analytical Methods. 2015;7(20):8889-95.

» 알레르기 혈액 검사의 종류_IgG

American Academy of Allergy Asthma & Immunology

Allergist Resources ⌄ Practice Management ⌄ Professional Education ⌄ Conditions & Treatments ⌄

Tools For The Public / Conditions Library / Allergies / The Myth Of IgG Food Panel Testing

The Myth of IgG Food Panel Testing

It is very common for patients to fe
have food allergies or
food intolerances / sensitivities. T
labels are often used interchangea
important to understand though, t
very different than intolerances or
There are excellent materials on th
help you distinguish between the t
allergy, the body is making an imm
the food, and this can be dangerou
intolerance or sensitivity, the body
processing or digesting the food a
obviously be uncomfortable).

불행히도 이 답을 줄 수 있는 단일 테스트는 없습니다. 식품 민감성을 진단할 수 있다고 주장하고 일반적으로 사용할 수 있는 검사는 식품 IgG 검사입니다. 다양한 회사에서 제공하는 이 검사는 여러 식품(단일 패널 검사로 일반적으로 90-100가지 식품)에 IgG 수치를 보고하며 IgG 수치가 높은 식품을 제거하면 여러 증상이 개선될 수 있다고 주장합니다. 일부 웹사이트에서는 이 검사를 사용하는 식단이 과민성 대장 증후군, 자폐증, 낭포성 섬유증, 류마티스 관절염 및 간질 증상에 도움이 될 수 있다고 보고하기도 합니다.

이 테스트는 보고된 작업을 수행할 수 있다고 과학적으로 입증된 적이 없다는 것을 이해하는 것이 중요합니다. 이 검사의 사용을 뒷받침하기 위해 제공되는 과학적 연구는 종종 구식이고, 평판이 좋지 않은 저널에 게재되어 있으며, 많은 사람들이 문제의 IgG 검사를 사용하지도 않았습니다. IgG의 존재는 음식 노출에 대한 면역 체계의 정상적인 반응일 가능성이 높습니다. 사실, 식품에 대한 IgG4 수치가 높은 것은 단순히 해당 식품에 대한 내성과 관련이 있을 수 있습니다.

사용을 뒷받침할 증거가 부족하기 때문에 다음을 포함한 많은 조직에서 미국 알레르기, 천식 및 면역학 학회 그리고 캐나다 알레르기 및 임상 면역학 학회(Canadian Society of Allergy and Clinical Immunology) 식품 알레르기 또는 식품 과민증/민감성을 진단하기 위해 IgG 검사를 사용하지 말 것을 권장했습니다.

Allergy EUROPEAN JOURNAL OF ALLERGY AND CLINICAL IMMUNOLOGY EAACI

Position Paper | Free Access

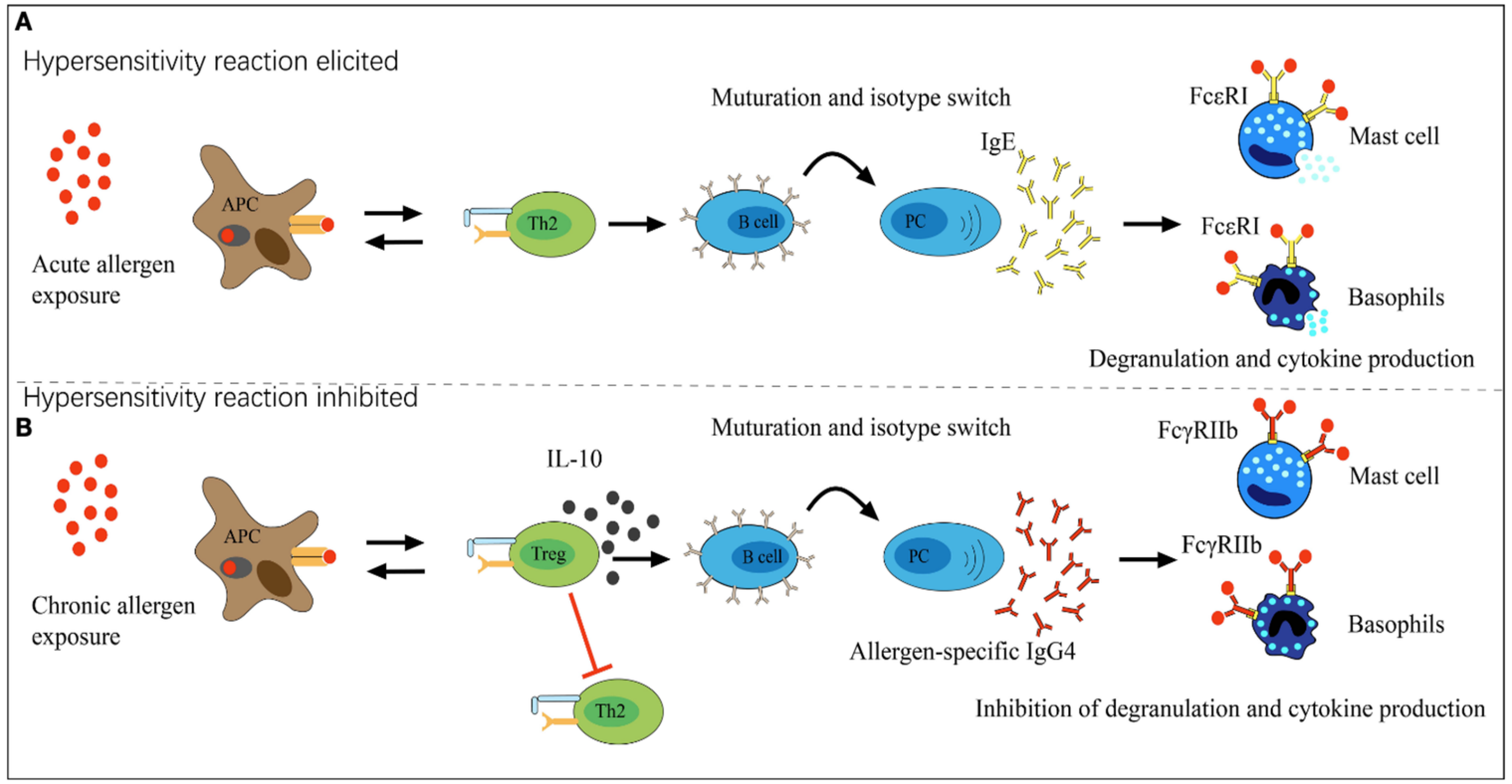
EAACI Food Allergy and Anaphylaxis Guidelines: diagnosis and management of food allergy

A. Muraro, T. Werfel, K. Hoffmann-Sommergruber, G. Roberts, K. Beyer, C. Bindslev-Jensen, V. Cardona, A. Dubois, G. duToit, P. Eigenmann, M. Fernandez Rivas, S. Halken ... See all authors ⌄

(F) Unconventional tests, including specific IgG testing

There are no unconventional tests which can be recommended as an alternative or complementary diagnostic tool in the workup of suspected food allergy, and their use should be discouraged	III	C	(48)

IgG



IgG4는 면역 관용의 지표

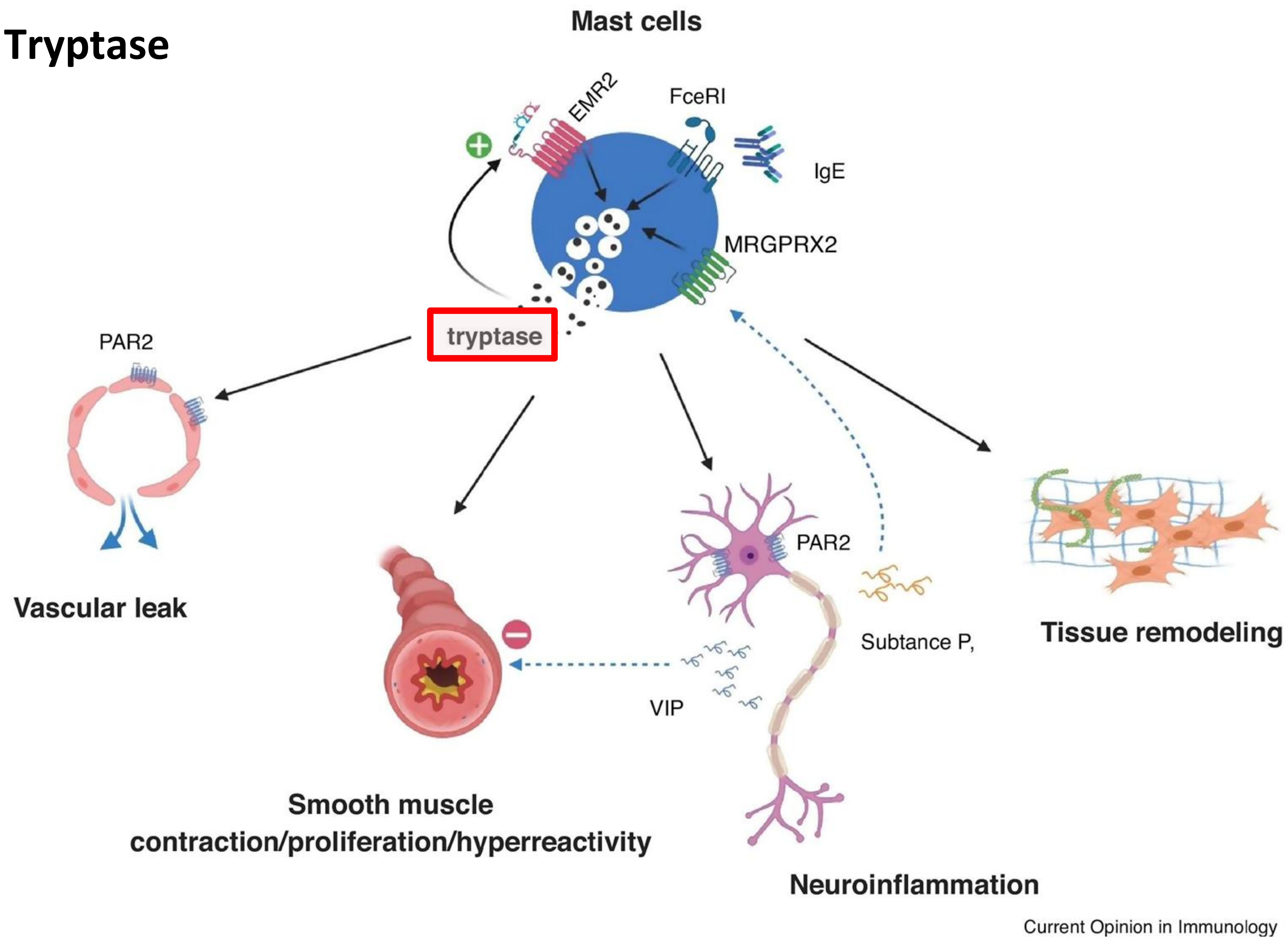
반복 항원 노출 시 증가, 임상 증상/IgE와 상관 없음

알레르기내과에서 IgG4를 활용하는 경우

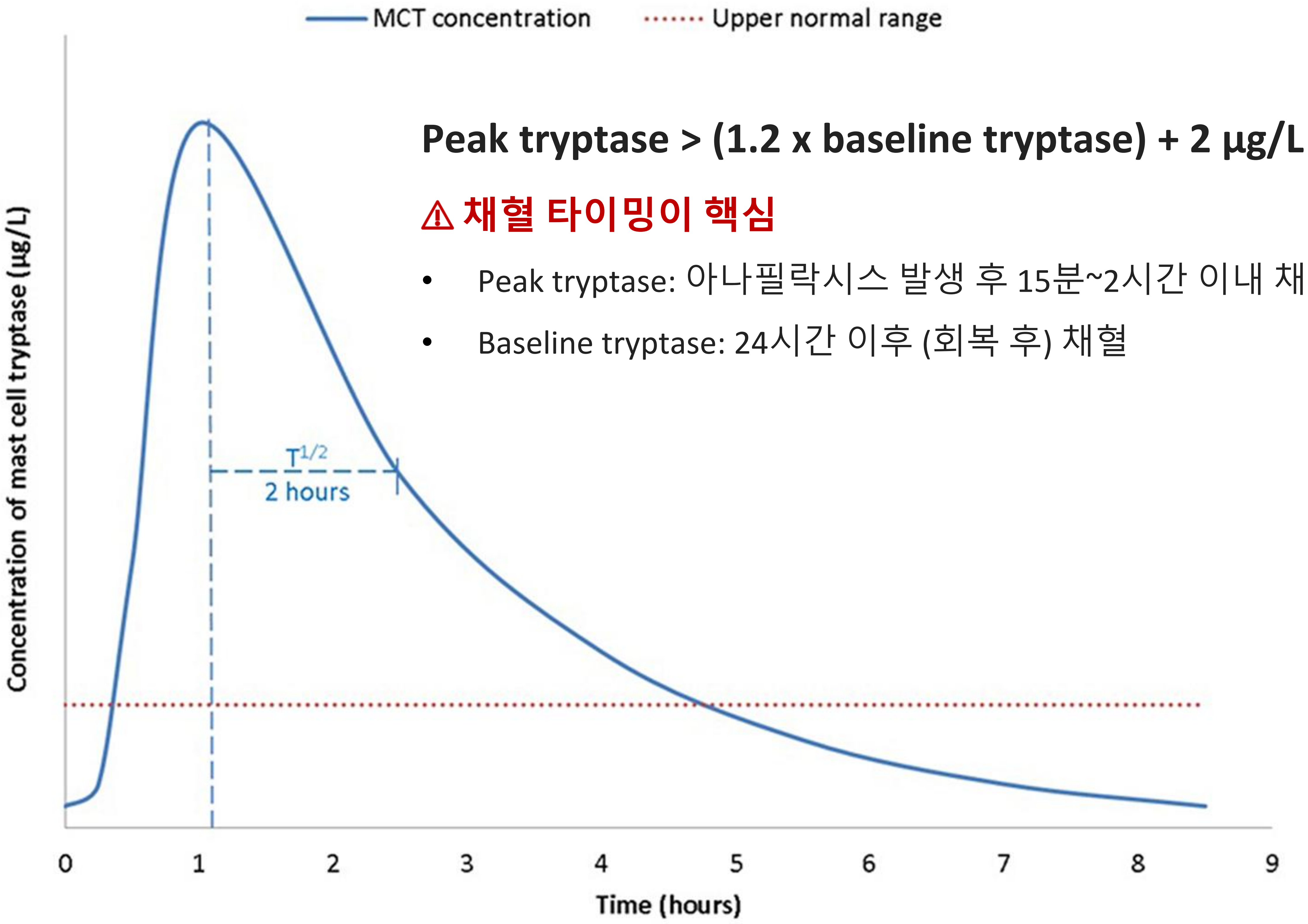
- 알레르겐 면역치료(AIT) 효과 모니터링 등

Stapel SO et al. Allergy. 2008 Jul;63(7):793-6.
AAAI Board of Directors. J Allergy Clin Immunol. 1995 Mar;95(3):652-4.
Wüthrich B. Investig Allergol Clin Immunol. 2005;15(2):86-90.
Qin L et al. Front Immunol. 2022 Oct 25;13:1032909.
Muraro A et al. Allergy. 2014 Aug;69(8):1008-25.

Tryptase



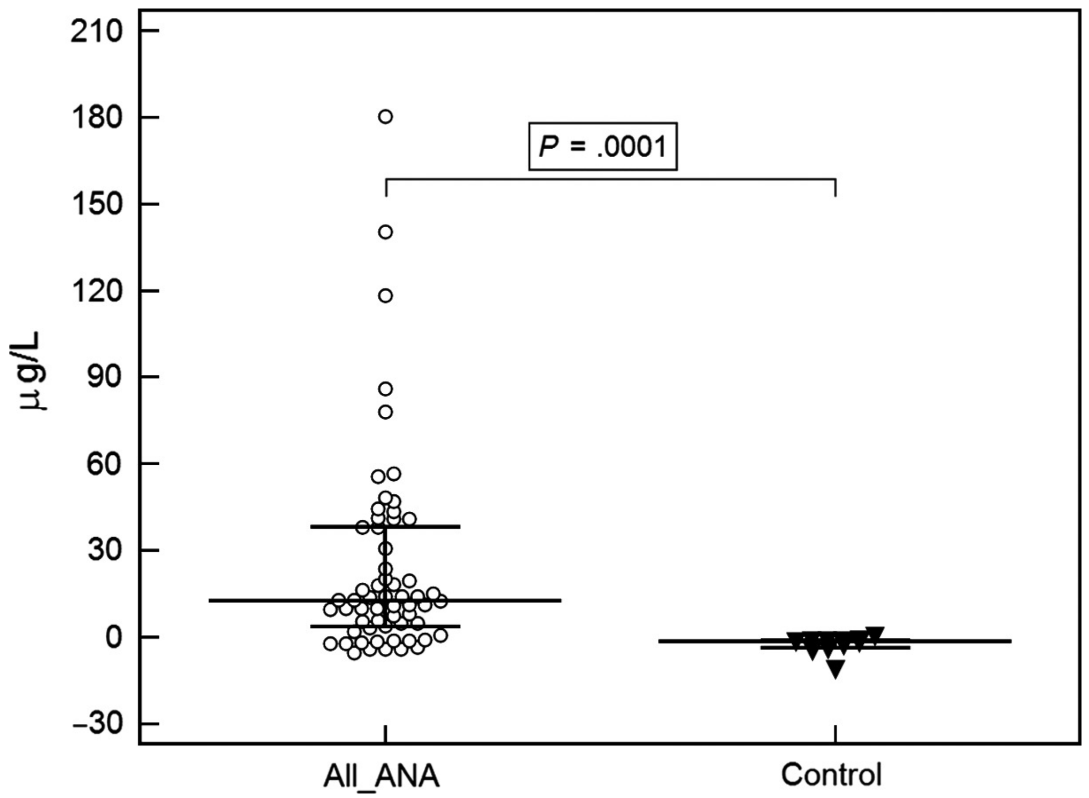
Lyons JJ, Yi T. Curr Opin Immunol. 2021 Oct;72:94-106.



Peak tryptase > (1.2 x baseline tryptase) + 2 µg/L

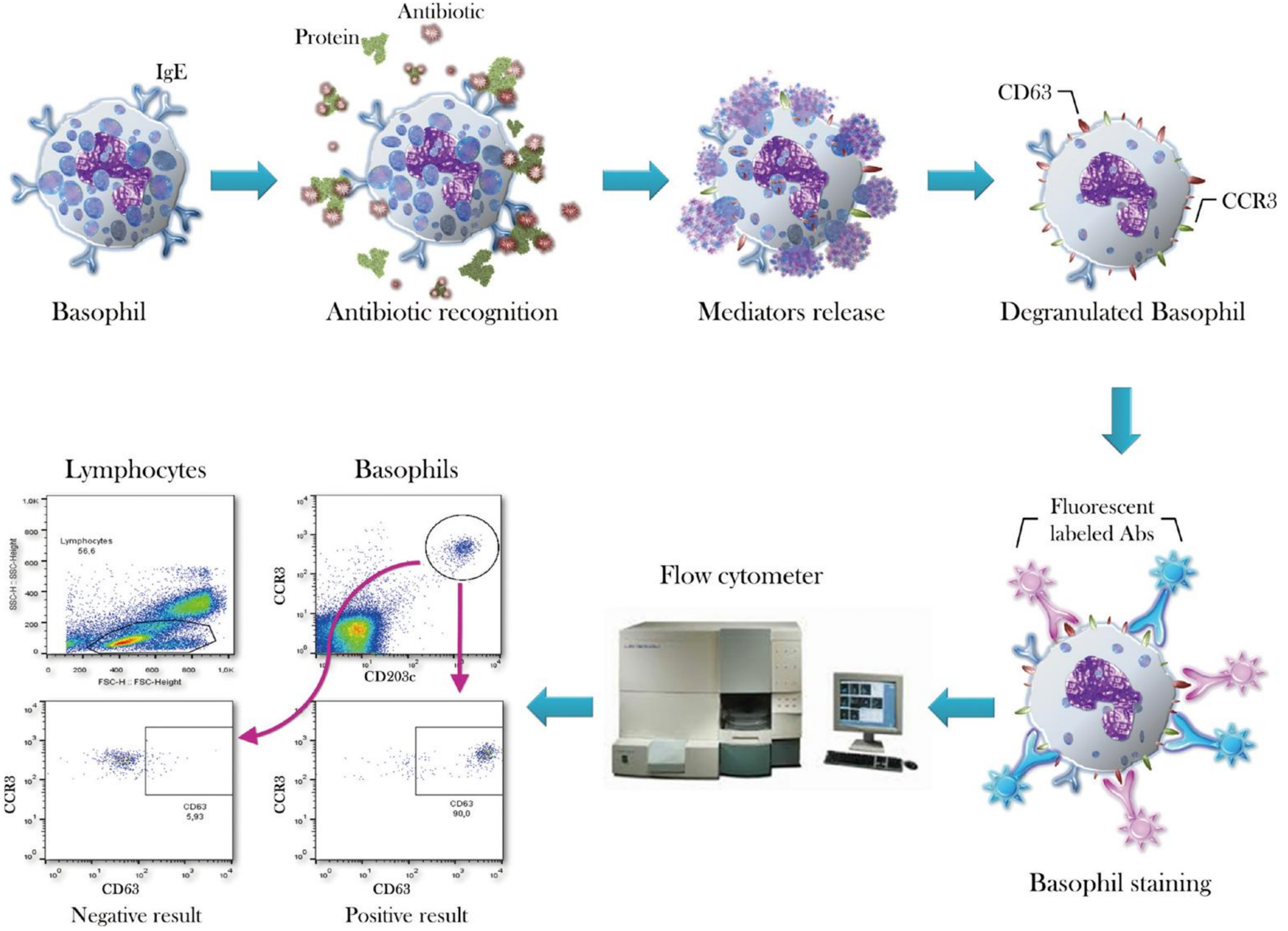
⚠ 채혈 타이밍이 핵심

- Peak tryptase: 아나필락시스 발생 후 15분~2시간 이내 채혈
- Baseline tryptase: 24시간 이후 (회복 후) 채혈



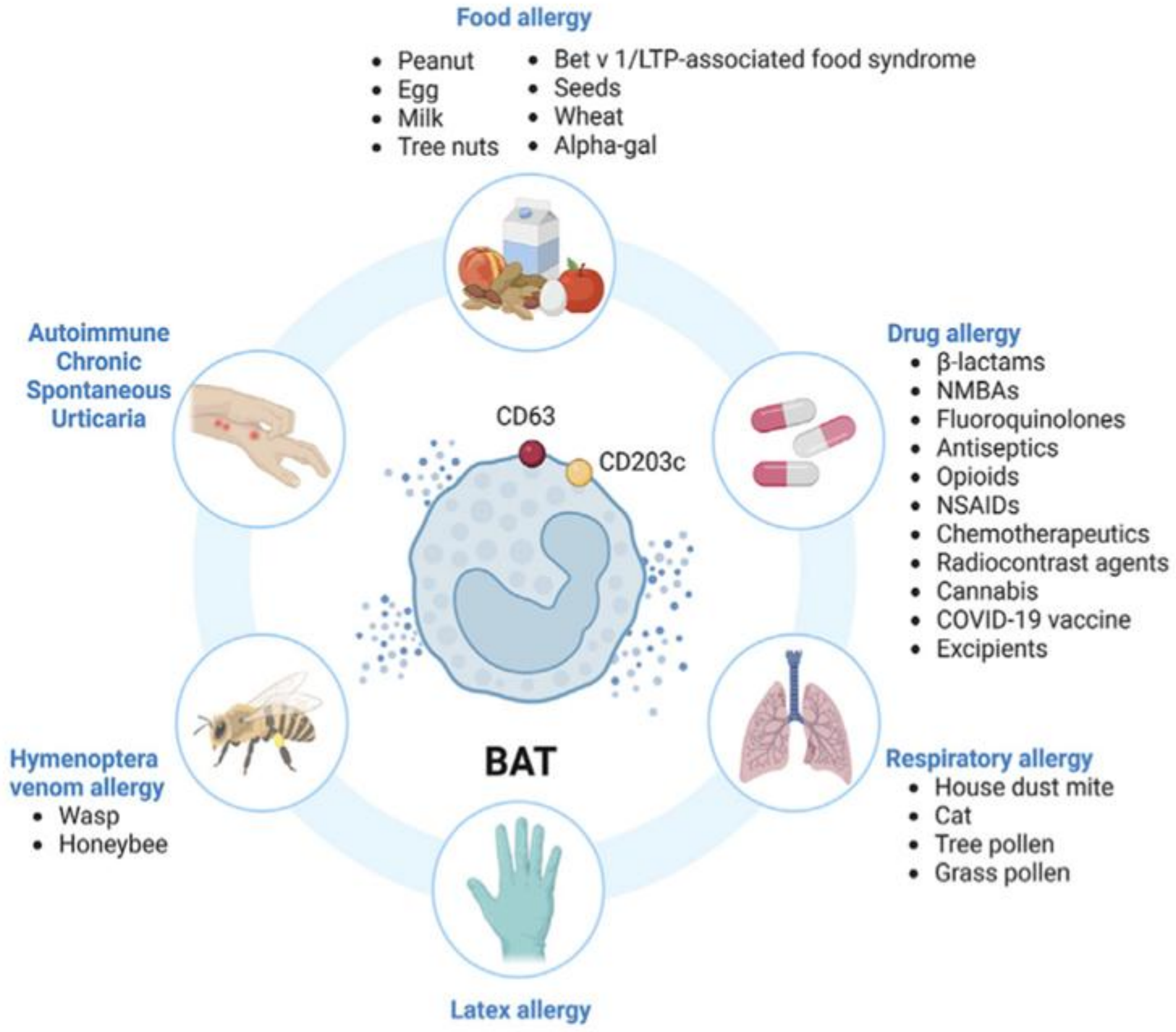
Beck SC et al. Front Immunol. 2019 Apr 5;10:494.
Baretto RL et al. Allergy. 2017 Dec;72(12):2031-2034.
Valent P et al. Int Arch Allergy Immunol.2012;157:215-225.

BAT
(Basophil Activation Test)

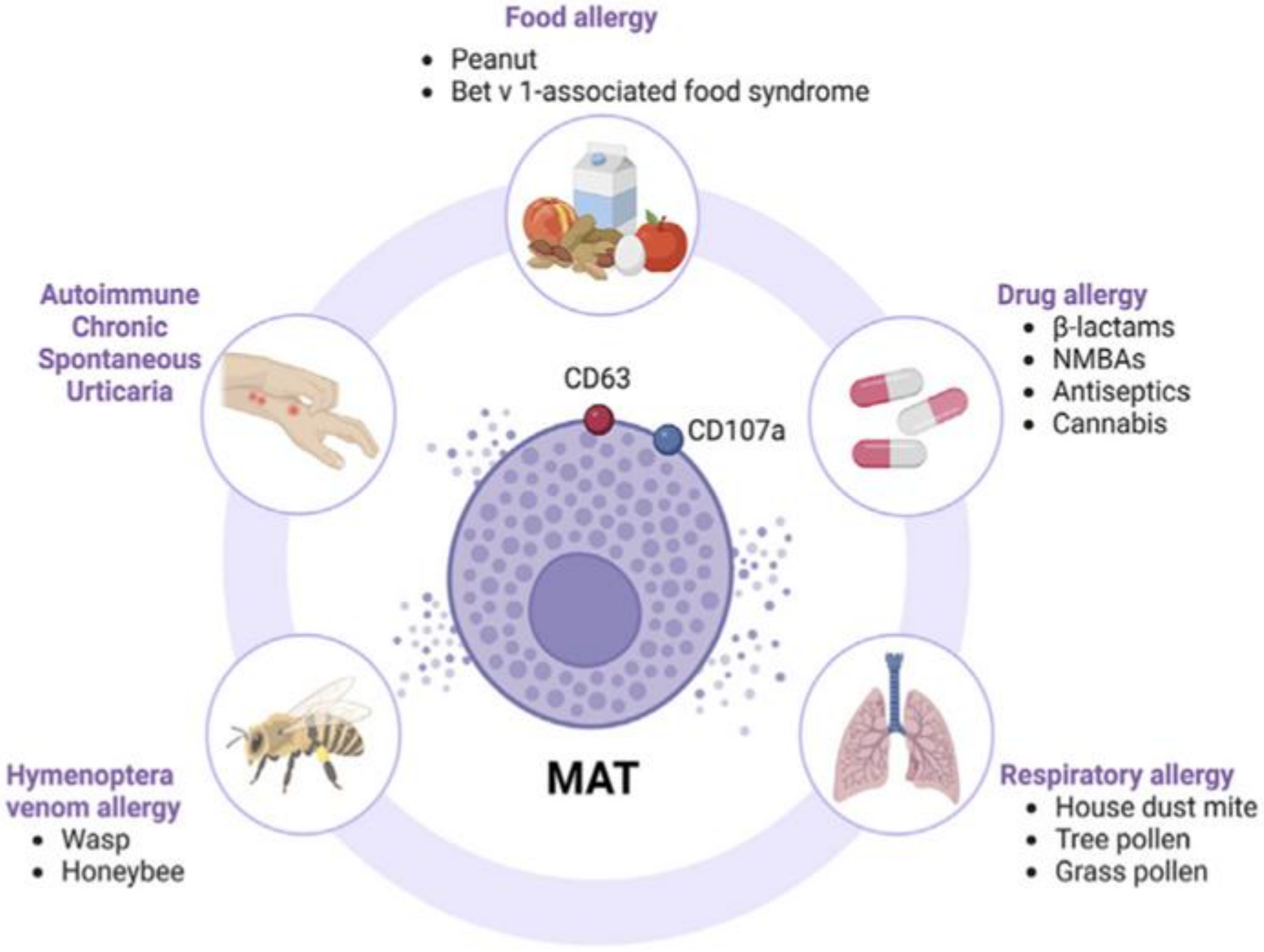


Doña I et al. Allergy Asthma Immunol Res. 2017;9(4):288-298.

» 알레르기 혈액 검사의 종류_MAT

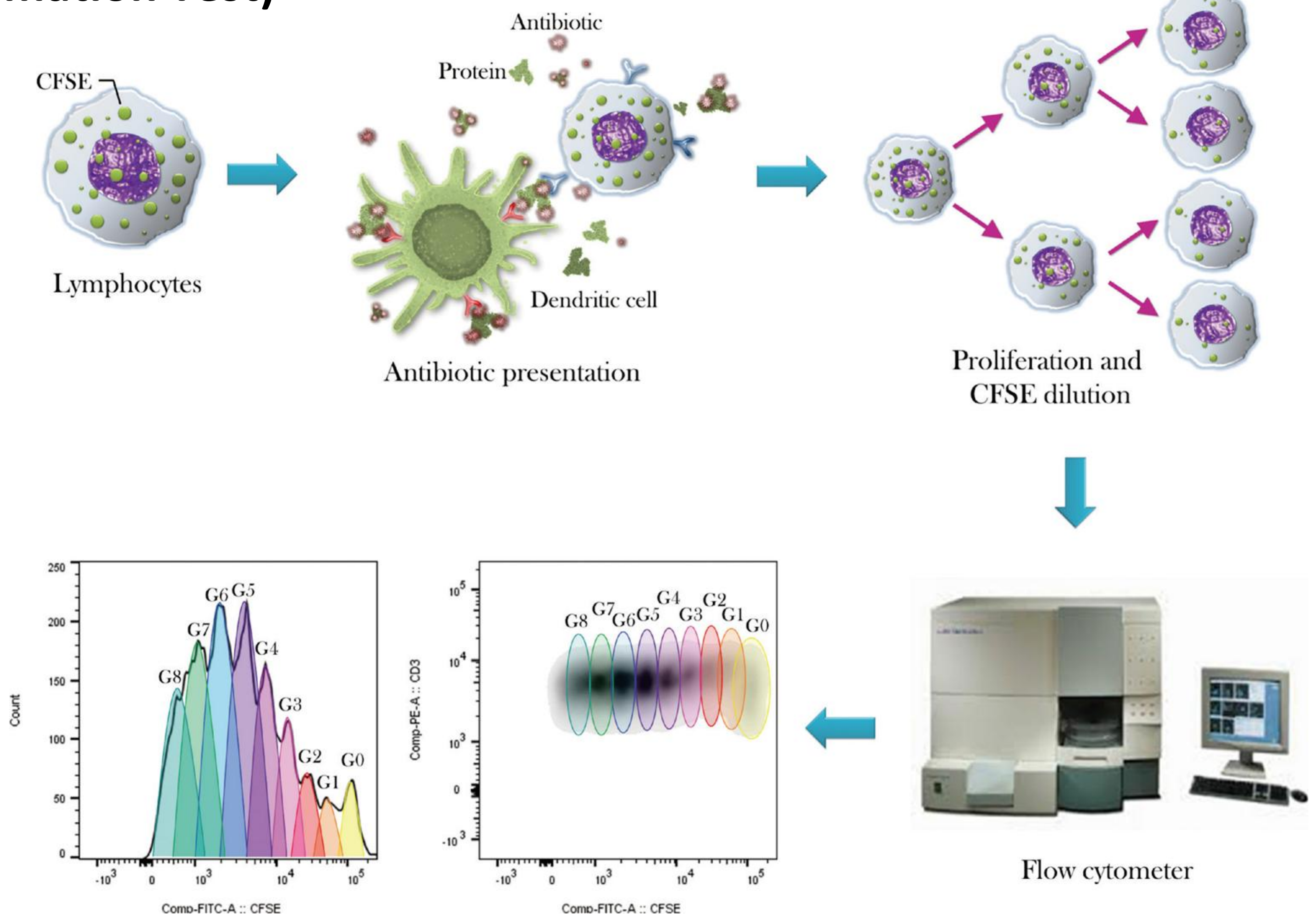


Mast cell Activation Test



Ebo DG et al. J Allergy Clin Immunol. 2025 Feb;155(2):286-297.

LTT
(Lymphocyte Transformation Test)



Doña I et al. Allergy Asthma Immunol Res. 2017;9(4):288-298.

Eosinophil

Peripheral eosinophilia (total eosinophil count)

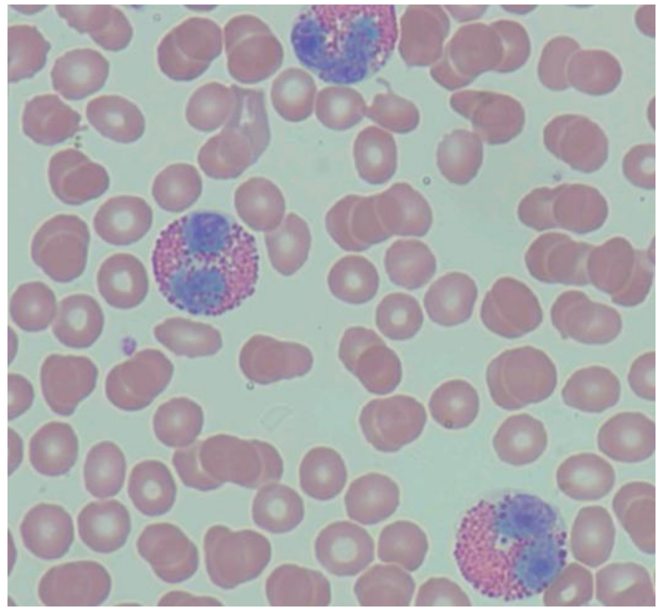
경증 500 – 1,500 cells/μL

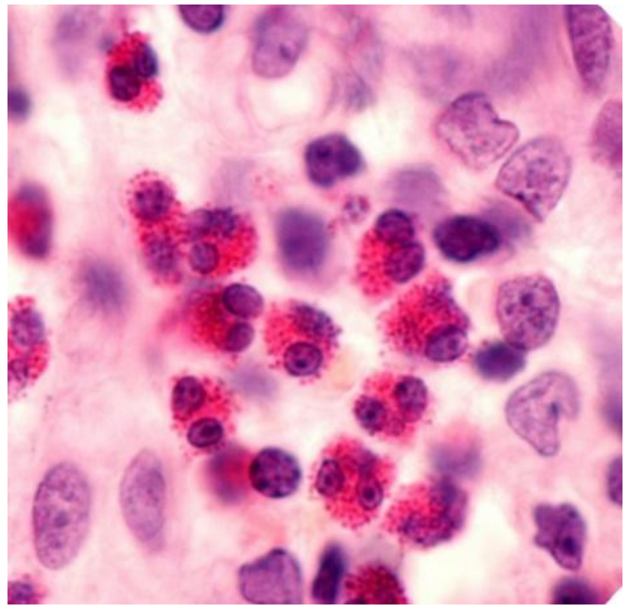
중등증 1,500-5,000 cells/μL

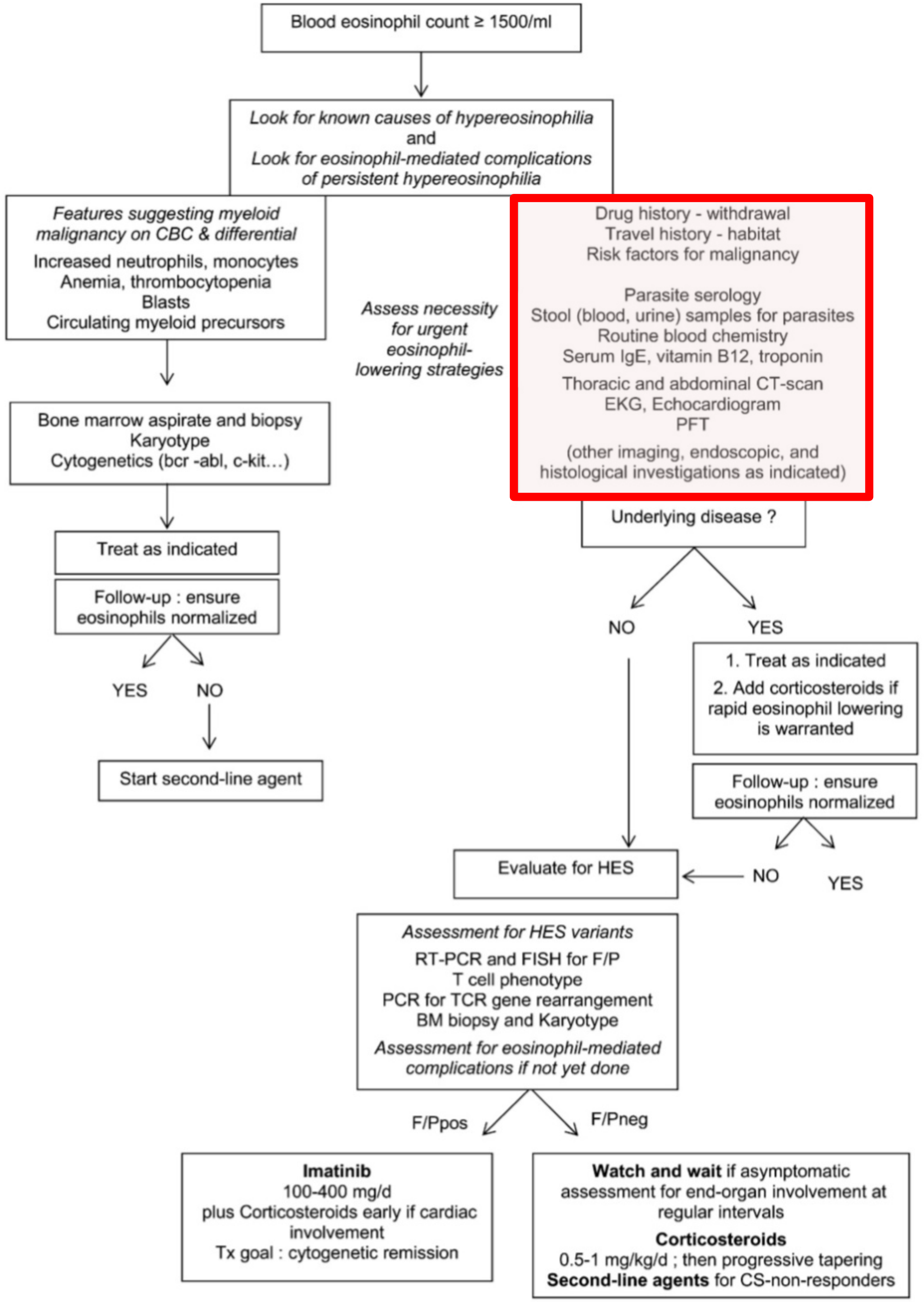
중증 > 5,000 cells/μL

Induced sputum eosinophil count

≥ 3% in type 2 inflammation (good response to ICS)

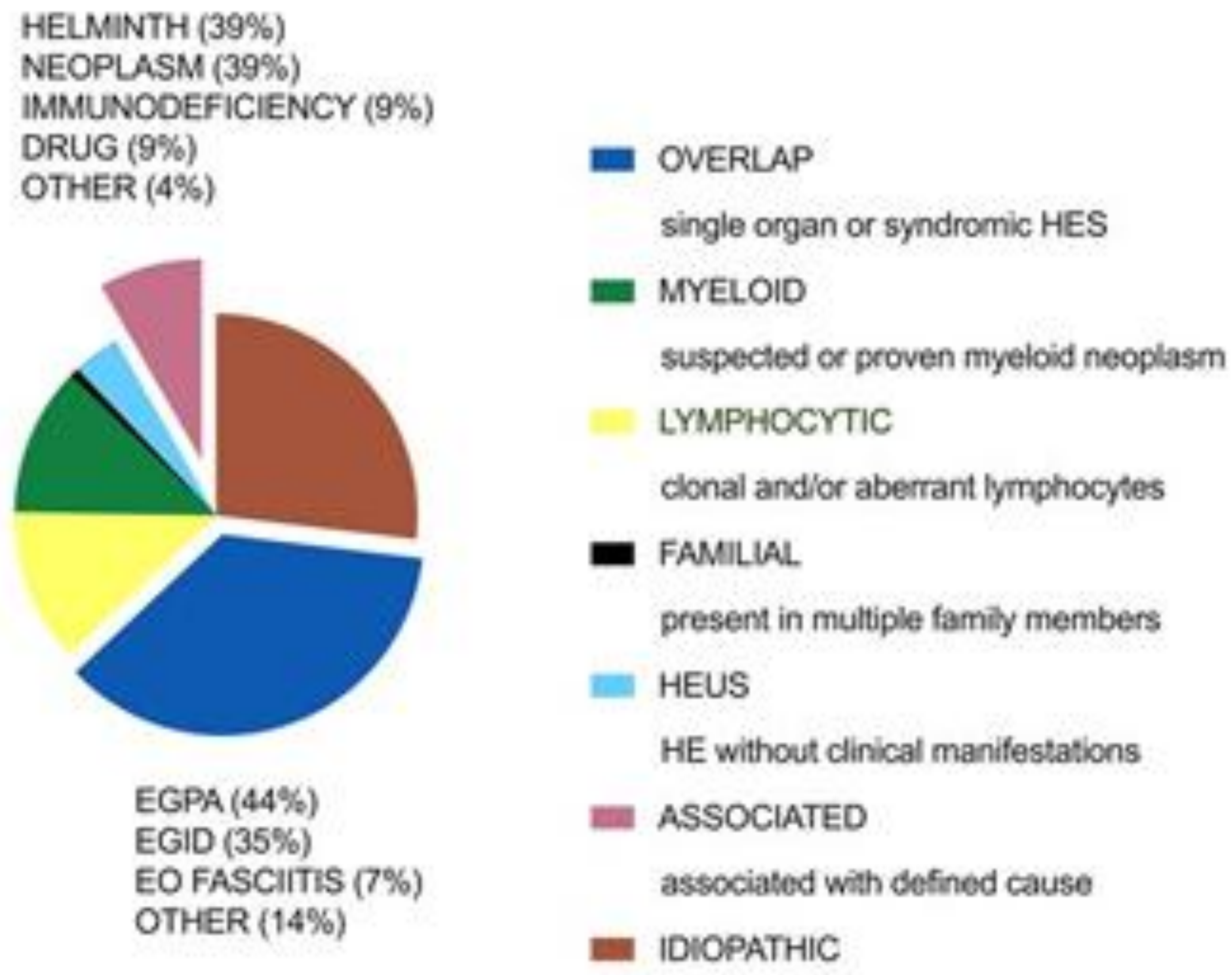


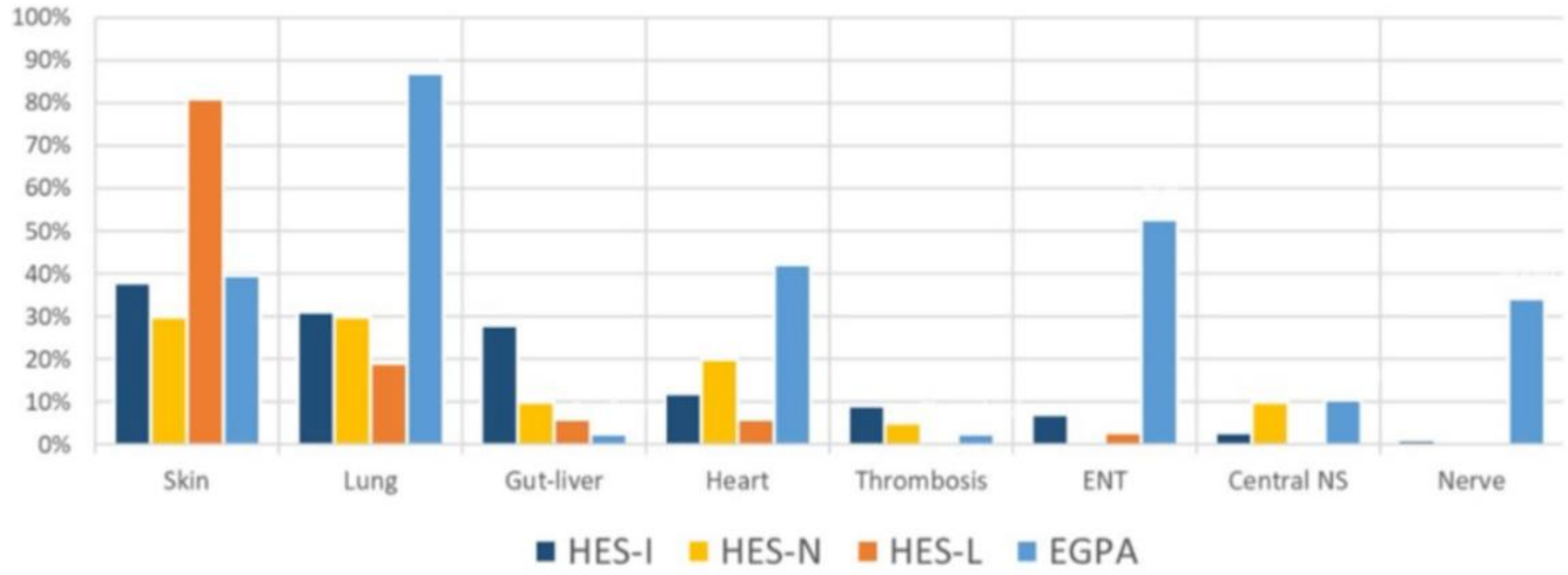




Roufosse F et al. J Allergy clin Immunol. 2010 Jul;126(1):39-44.

≫ 알레르기 혈액 검사의 종류_Eosinophil





Lefèvre G et al. Allergy. 2025 Apr;80(4):1100-1110.

» 알레르기 혈액 검사의 해석

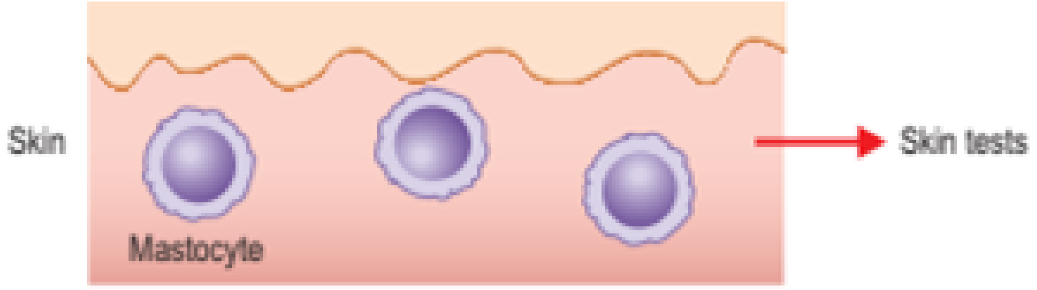
Allergen 감작 ≠ Allergy 진단

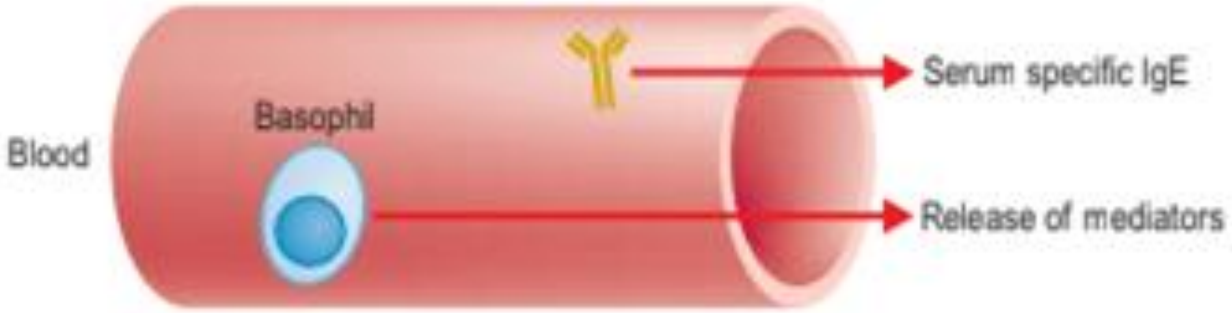
알레르기 진단 = 병력 AND 검사

- Allergen 특이 IgE 양성 (= 감작 되어있다)
- Allergen 노출 시 증상 재현 / 회피 시 소실
    - 실내/실외, 계절, 직업 등

» 알레르기 혈액 검사의 해석_SPT vs. Serologic test

Skin test	Serologic test
검사자 의존적	Automatic method (표준화/재현성)
30분 이내	수일
알레르기 반응 재현 가능	안전
피부질환, 약물 복용 제한	제한 없음
검사자가 항원 선택 가능 (crude extract only)	제품화
최대 55종 (Limited by space)	ImmunoCAP: 260 tests(한국) MAST : 55-176종





알레르기 혈액 검사의 해석_MAST vs. ImmunoCAP

MAST	ImmunoCAP
동시에 여러 알레르겐 검사 가능	**검사 당 1개**의 알레르겐 -> 의심항원을 선택적으로 검사
반정량적 : **class 0-6**	정량적 (kU/L) (양성 기준치 **≥ 0.35 kU/L**)
민감도, 특이도가 상대적으로 낮다	흡입 항원 : 민감도/특이도 좋음 식품 항원 : 민감도 좋음, 특이도 나쁨
선별검사 (screening)	**표준/정밀 검사** **추적 모니터링 가능** (면역치료, 식품 알레르기 등)
흡입 알레르겐, 식품 알레르겐 최근 성분 항원도 추가	흡입 항원 (집먼지진드기, 꽃가루, 동물비듬, 곰팡이 등) 식품 항원 (계란, 우유, 견과류, 갑각류, 밀가루/곡류 등) 곤충독 (꿀벌독, 말벌독, 불개미독 등) 약물 (페니실린, 세파클러 등) 성분 항원
패널 기반 비용 책정 (제품 따라 상이)	**항목당 비용**이 책정되어 개수가 늘어날수록 비용 증가 6종 급여 (6세미만 소아/Anaphylaxis 시 등 최대 12종)

알레르기 혈액 검사의 해석_MAST vs. ImmunoCAP

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journal homepage: www.elsevier.com/locate/clinchim

Specific IgE measurement using AdvanSure® system: Comparison of detection performance with ImmunoCAP® system in Korean allergy patients

Jae-Hyun Lee [a,b], Kyung Hee Park [a,b], Hyon-Suk Kim [c], Kyung Won Kim [b,d], Myung Hyun Sohn [b,d], Chang-Hoon Kim [b,e], Joo-Shil Lee [f], Chein-Soo Hong [a,b], Jung-Won Park [a,b,*]

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[c] Department of Laboratory Medicine, Yonsei University College of Medicine, Seoul, Republic of Korea
[d] Department of Pediatrics, Yonsei University College of Medicine, Seoul, Republic of Korea
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https://doi.org/10.3349/ymj.2017.58.4.786
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Comparison of the ImmunoCAP Assay and AdvanSure™ AlloScreen Advanced Multiplex Specific IgE Detection Assay

Kyung Hee Park[1,2], Jongsun Lee[2], Sang Chul Lee[1], Young Woong Son[1], Da Woon Sim[1,2], Jae-Hyun Lee[1,2], and Jung-Won Park[1,2]

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[2]Institute of Allergy, Yonsei University College of Medicine, Seoul, Korea.

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Performance of the PROTIA™ *Allergy-Q*® System in the Detection of Allergen-specific IgE: A Comparison With the ImmunoCAP® System

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RESEARCH ARTICLE
WILEY

Inter-laboratory comparison of semiquantitative allergen-specific Immunoglobulin E test: 7 years of experience in Korea

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AJCP / Original Article

Evaluation of AdvanSure AlloScreen Max Panel With 92 Different Allergens for Detecting Allergen-Specific IgE

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Key Words: Multiple allergen simultaneous test; MAST; 92 different allergens; Allergen-specific IgE; Serum IgE concentration; AdvanSure AlloScreen Max; ImmunoCAP

Am J Clin Pathol June 2019;151:628-637

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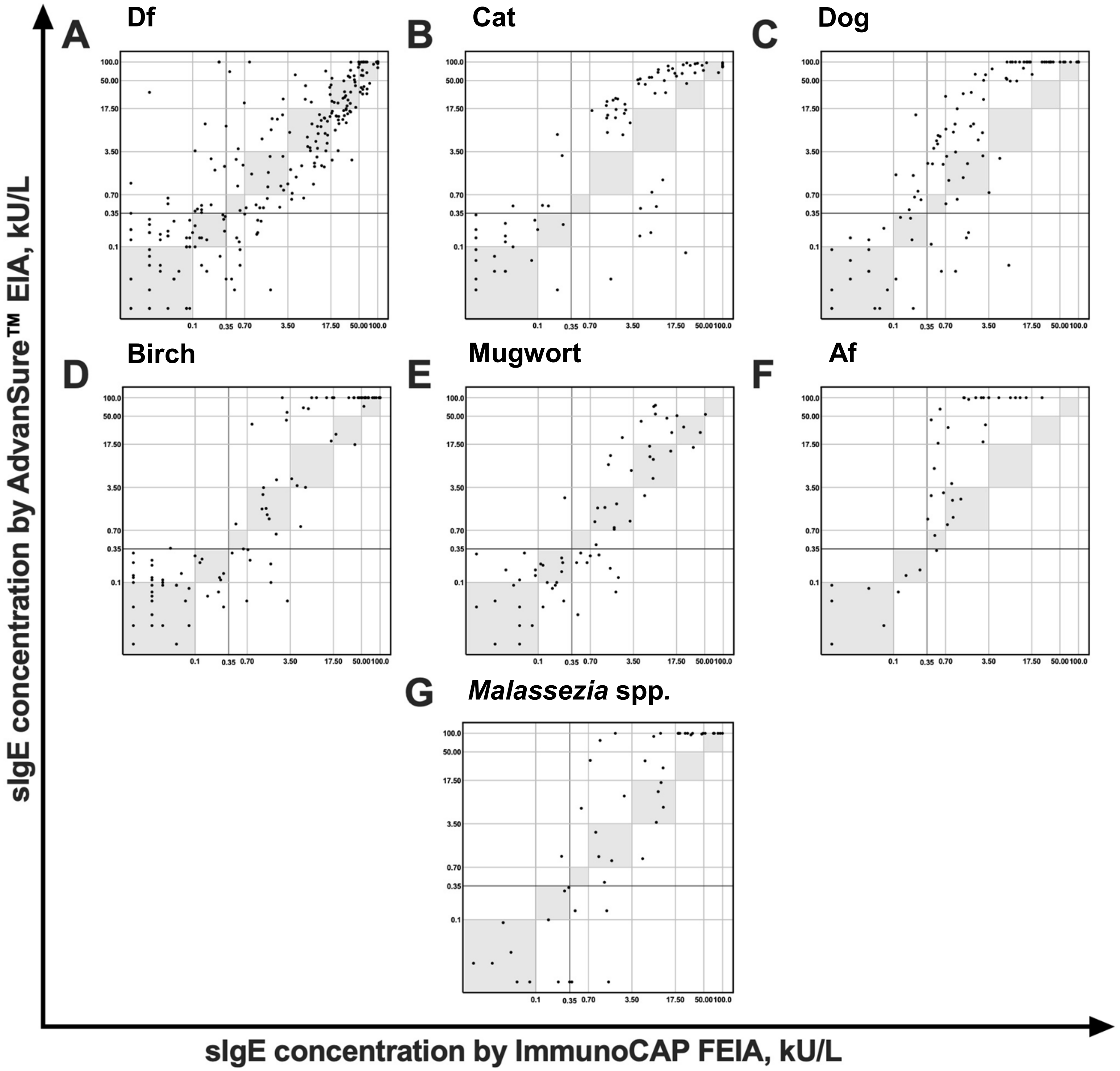
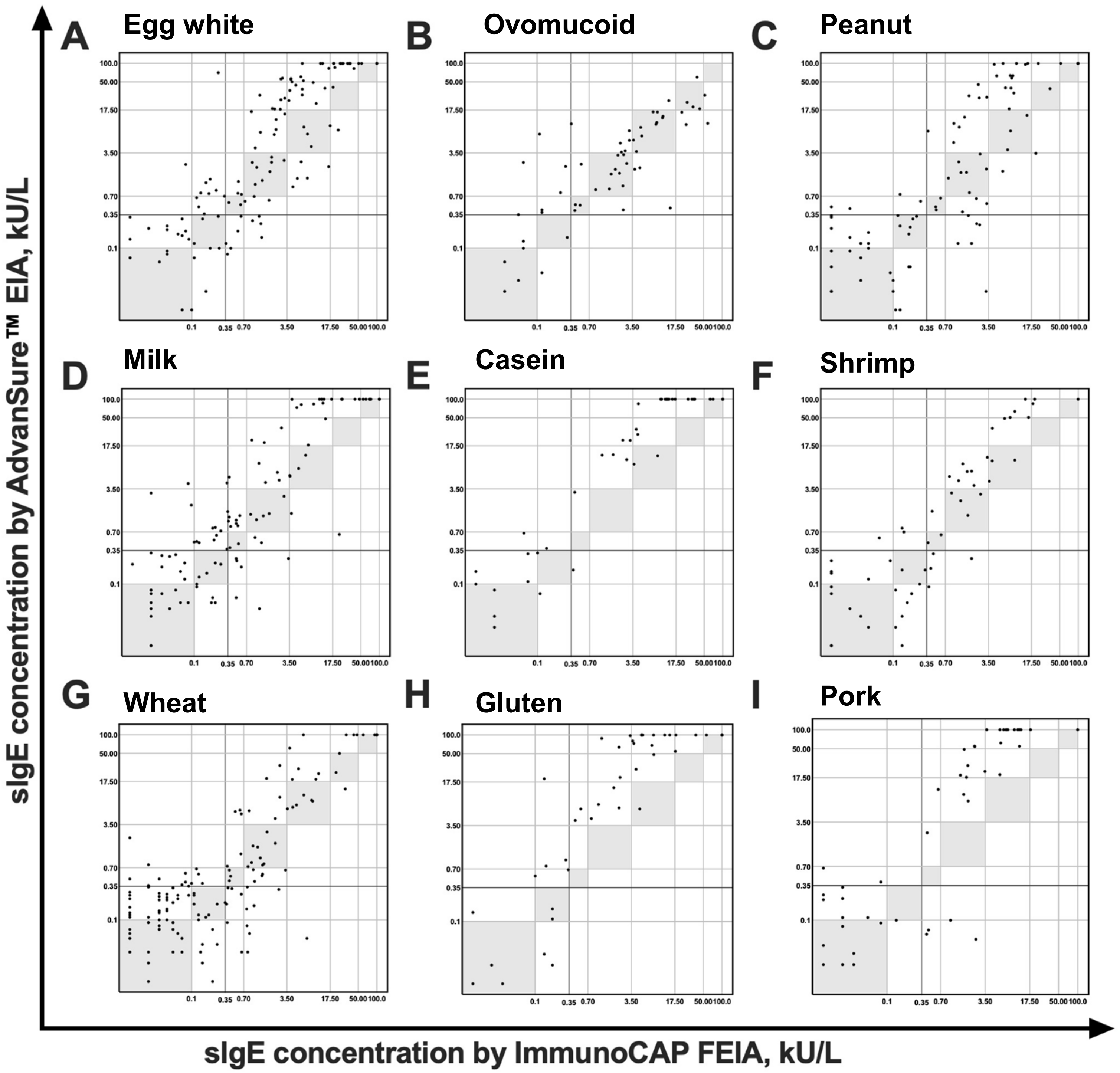
알레르기 혈액 검사의 해석_MAST vs. ImmunoCAP





≫ 알레르기 혈액 검사의 해석_임상과 검사의 불일치

⚠ 임상 맥락과 함께 해석이 필요한 경우

① 검사 양성 + 증상 없음

→ "감작(sensitization)"만 된 것

→ 노출 시 증상 없으면 임상적 알레르기 아님

② 검사 음성 + 증상 있음

→ IgE 비의존성 기전, Local IgE (예: local allergic rhinitis)

→ 검사한 항원 패널에 원인 항원이 없는 경우

③ Total IgE 높은데 특이 IgE 다 낮음

→ 기생충 감염, 비특이적 IgE 상승 가능

→ 검사한 항원 패널에 원인 항원이 없는 경우

④ Class/수치 낮은데 아나필락시스 발생

→ IgE 수치와 증상 중증도는 비례하지 않음

→ 비만세포 반응성, 항원 노출량, augmentation factor 등이 관여

⑤ CAP/MAST 결과가 서로 다름

→ Allergen extract 표준화 차이, solid phase / detection system 차이

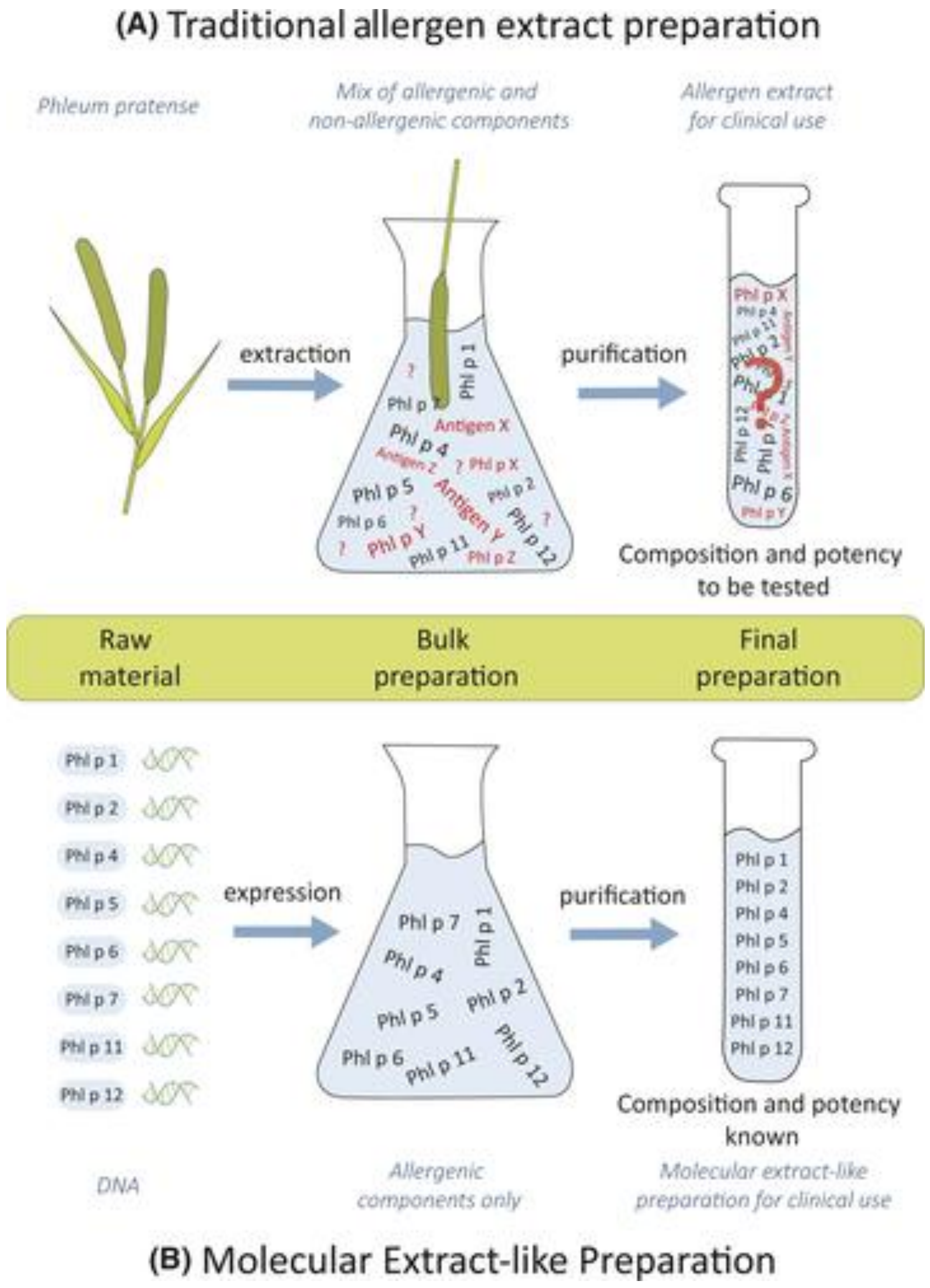
→ Crude extract vs. component → 식품 알레르겐에서 특히 차이 크다

⑥ 식품 먹어도 괜찮은데 특이 IgE 양성

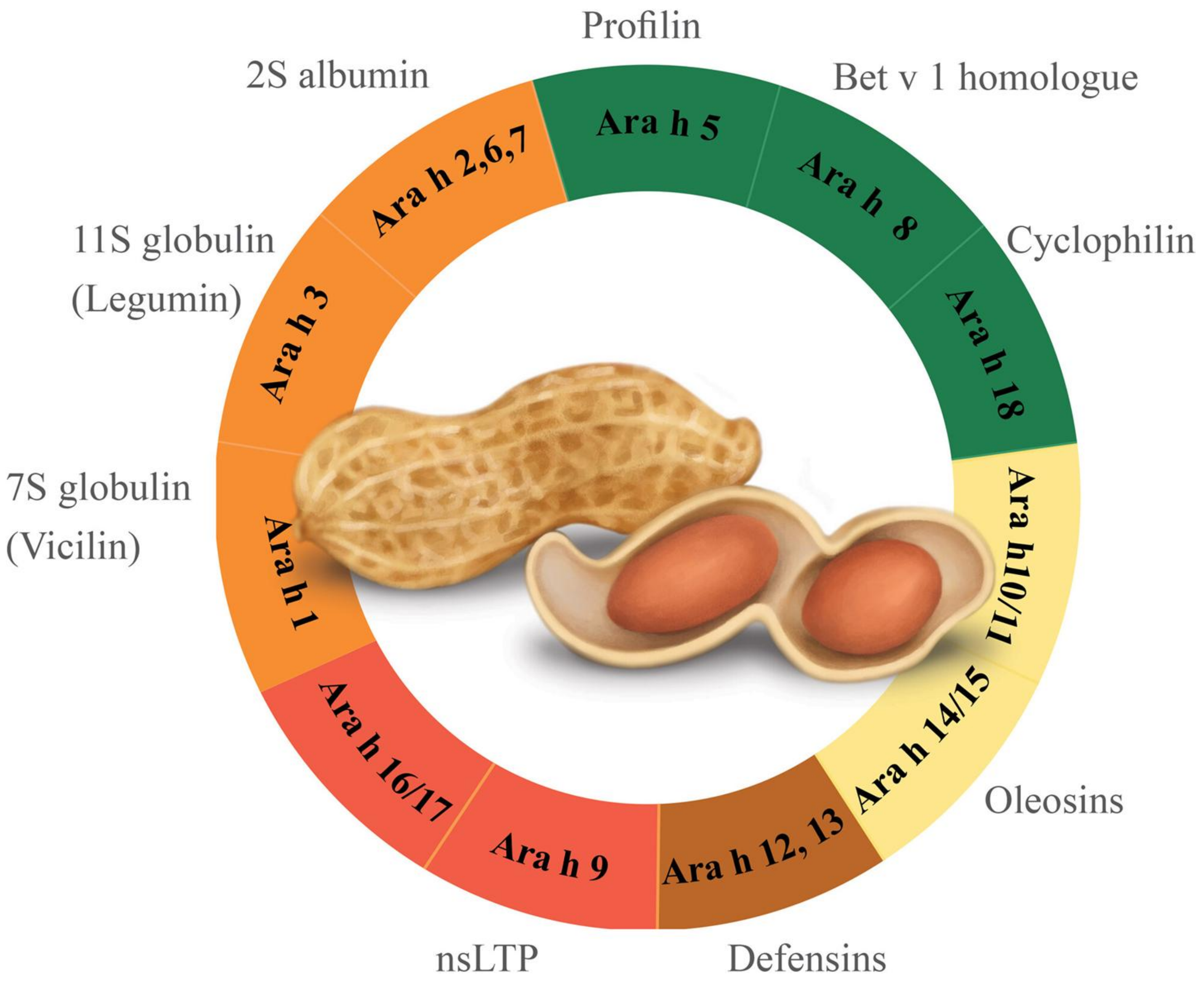
→ 식품 항원은 교차감작이 흔하다

→ PFAS: 꽃가루 교차반응, 열처리 시 증상 없음

→ CRD (component)로 PR-10, profilin 확인 시 PFAS 구분 가능



Component Resolved Diagnostics
Component allergen, 성분 항원



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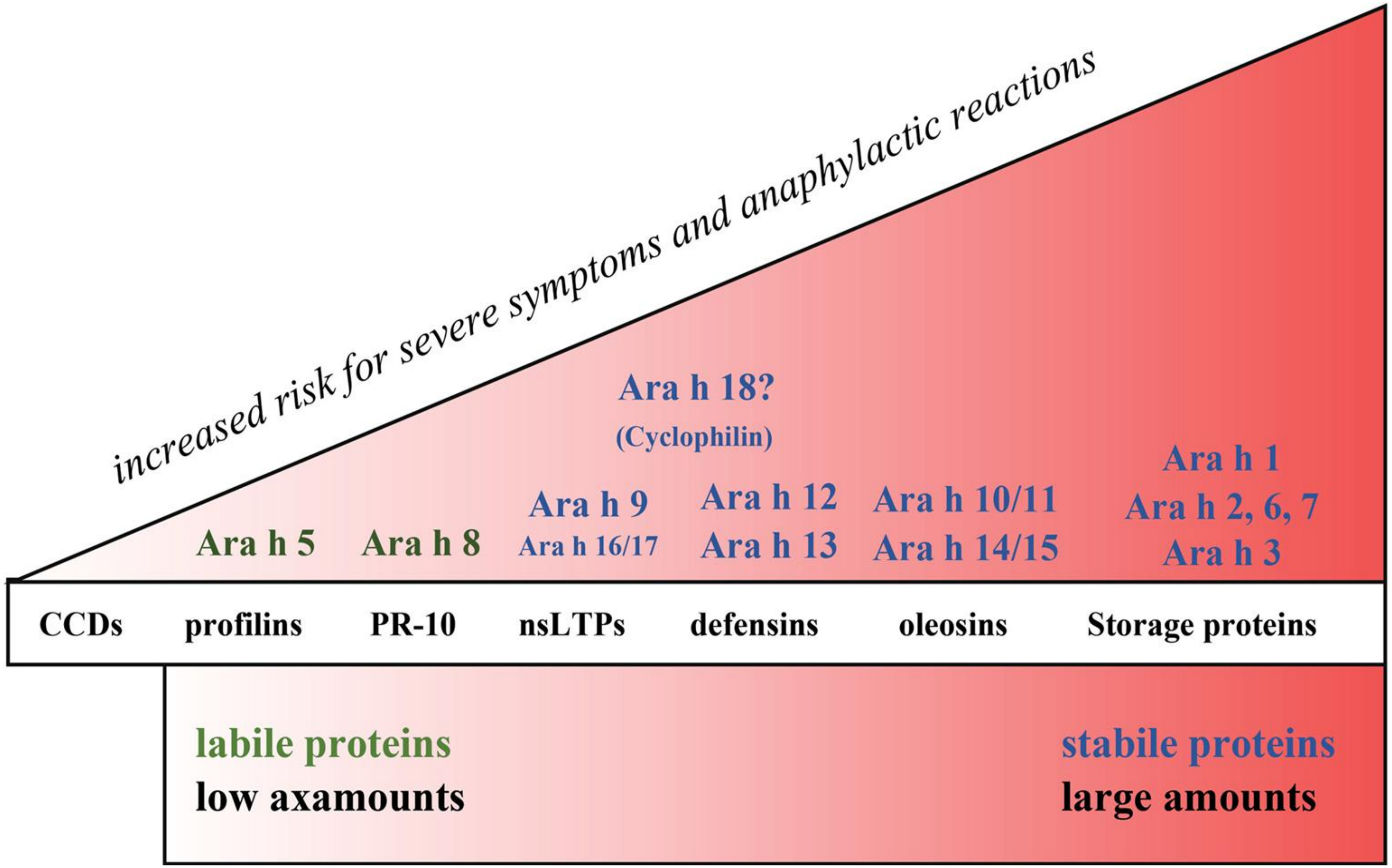
» 알레르기 혈액 검사의 해석_CRD



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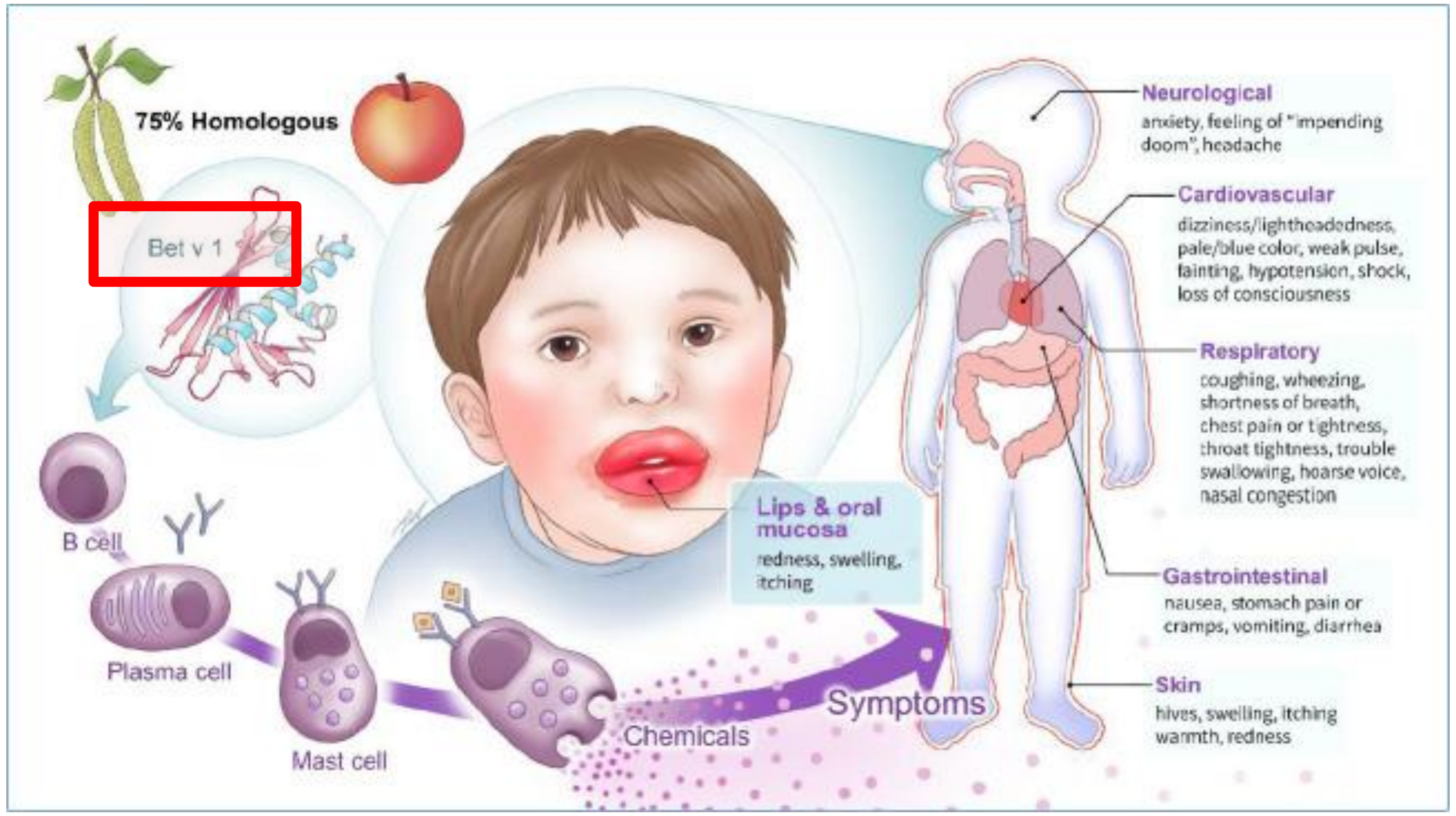
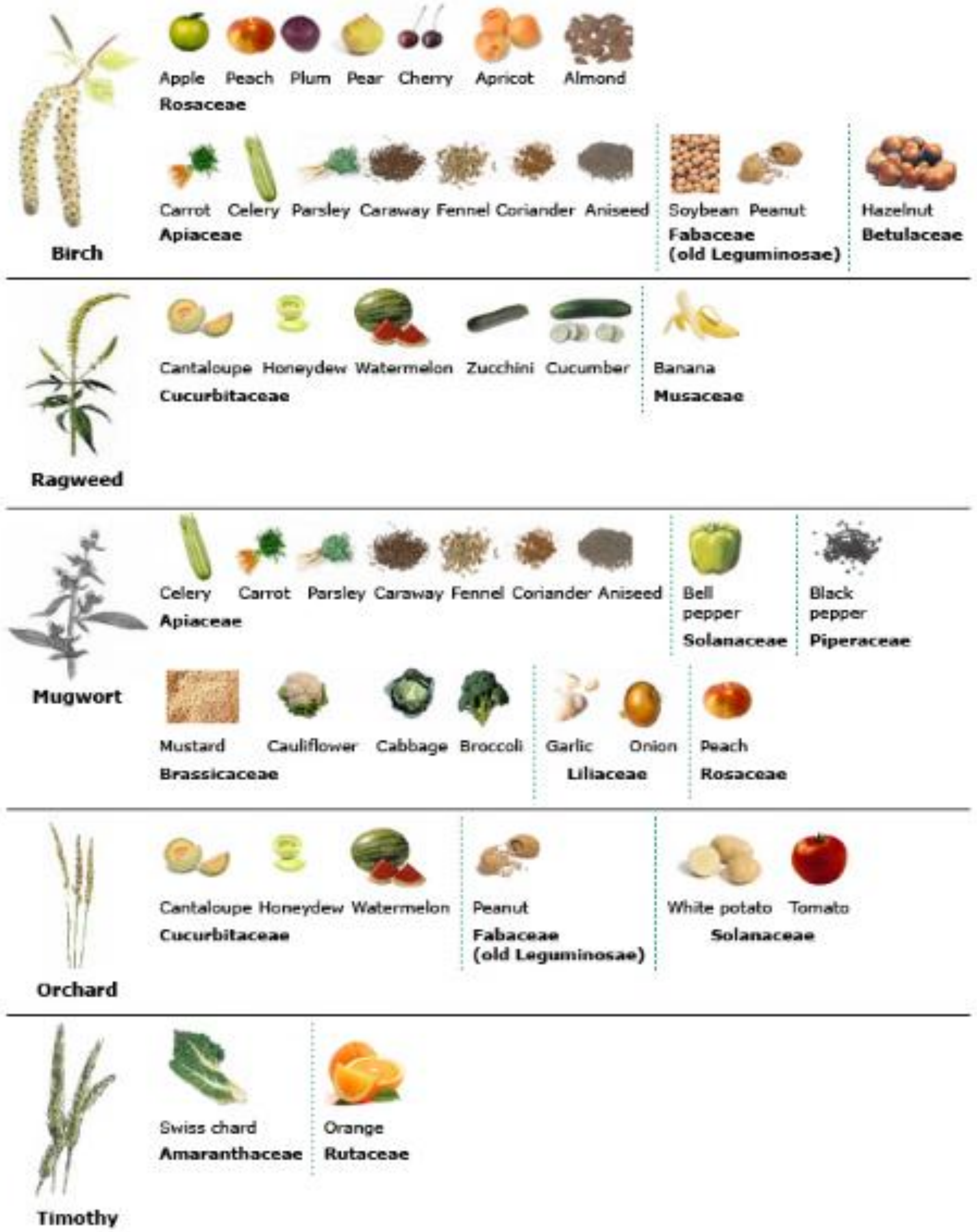
» 알레르기 혈액 검사의 해석_PFAS

PFAS
(Pollen-Food Allergy Syndrome)





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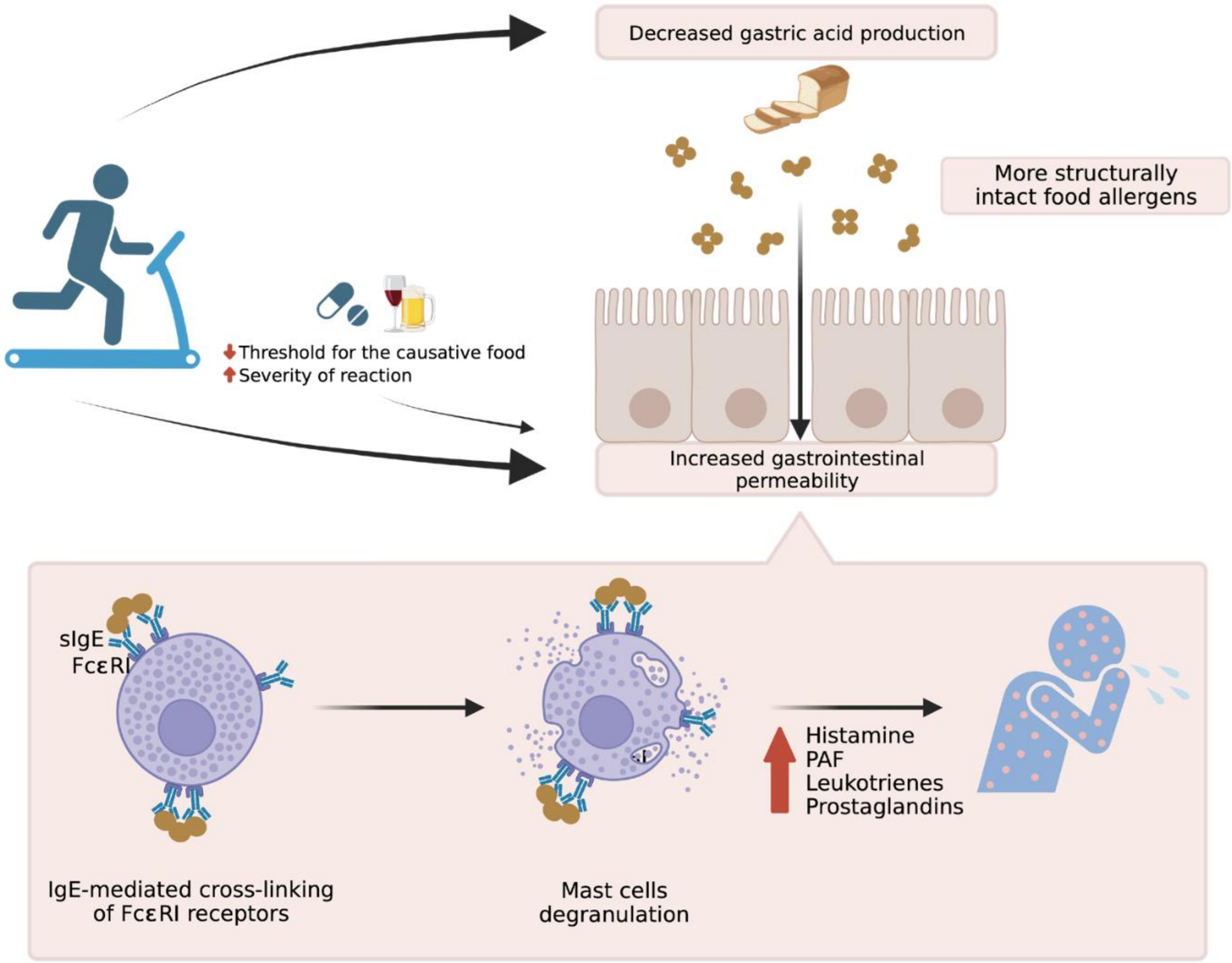

WDEIA
(Wheat Dependent Exercise-Induced Anaphylaxis)

Table. Clinical features of wheat allergy phenotypes

	Onset ages	Exposure route	Major allergens, provocation factors	Pathogenesis	Diagnosis
WDEIA	Mainly adults, but occasionally children	Oral, skin	Water-salt insoluble gluten (ω-5 gliadin, glutenins) with provoking factors (exercise, NSAIDs, alcohol)	IgE-mediated	SPT or sIgE to ω-5 gliadin, gluten. If needed, food challenge test
Immediate allergic reactions of wheat allergy	Mainly children	Oral	Water-salt soluble α-amylase inhibitors, lipid transfer proteins; and water-salt insoluble gluten (ω-5 gliadin, glutenins) without provoking factors	IgE-mediated	SPT or sIgE total extract of wheat, ω-5 gliadin, gluten. If needed, food challenge test
Atopic dermatitis with wheat sensitization	Neonates to children	Oral, skin	Water-salt soluble α-amylase inhibitors, lipid transfer proteins; water-salt insoluble gluten (gliadins, glutenins)	IgE and type 2 inflammation	SPT or sIgE to total extract of wheat, ω-5 gliadin, gluten. If needed, food challenge test
Eosinophilic esophagitis	Children to adults	Oral	Milk, wheat, egg, soy, fish, shellfish, peanut, tree nuts	Type 2 inflammation	Elimination diet
Baker's occupational asthma and rhinitis	Adults	Inhalation	Water-salt soluble α-amylase inhibitors, lipid transfer proteins, gliadins, aspergillus α-amylase, other cereals	IgE and type 2 inflammation	SPT or sIgE total extract of wheat, rye. Methacholine challenge test. If needed, bronchial provocation test

WDEIA, wheat-dependent exercise-induced anaphylaxis; NSAID, nonsteroidal anti-inflammatory drug; IgE, immunoglobulin E; SPT, skin prick test; sIgE, specific immunoglobulin E.

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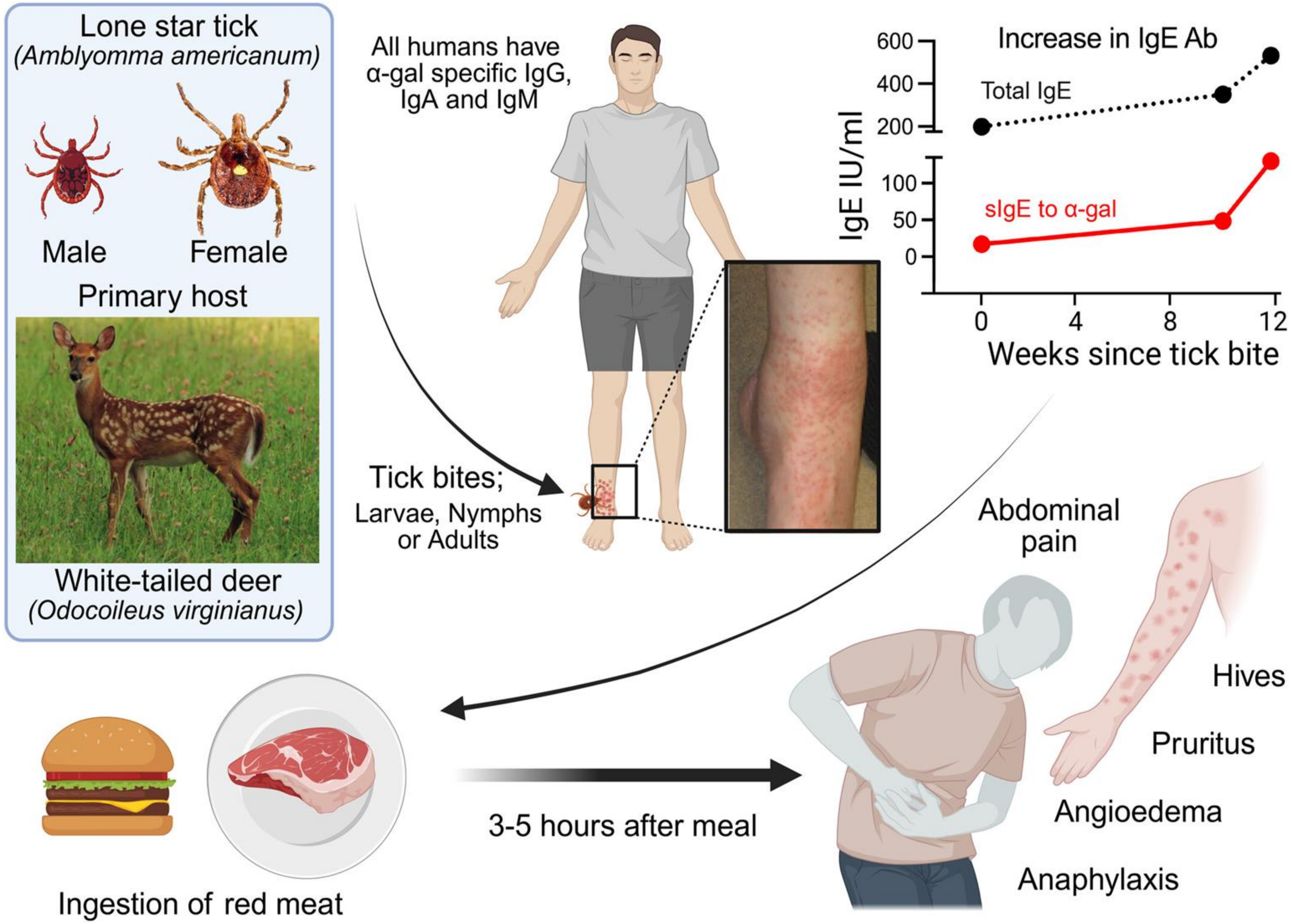
Galactose-α-1,3-galactose allergy
(Red meat syndrome)



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» Summary

감작 ≠ 알레르기

결과 해석 시 노출 환경과 임상 증상을 함께 고려해야한다

MAST는 다수 항원을 한 번에 확인하는 선별검사로 유용하다

ImmunoCAP은 의심 항원을 정량적으로 평가, 추적하는 데 적합하다

Tryptase, BAT, LTT 는 특수 상황에서 진단의 정확도를 높여준다

CRD 는 교차반응 해석, 위험도 평가, 회피 및 면역치료 계획 수립에 도움을 준다

경청해주셔서

감사합니다.

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